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Phase-Conditioned Implicit Neural Representations for Three-Dimensional Left Ventricular Reconstruction and Myocardial Wall Thickness Estimation

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Master in Artificial Intelligence

Supervisor:
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TECNOLOGIAS
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Department of Information Science and Technology

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To my grandparents, my parents, and my brother.

For your love, your patience, and the support that made this possible.

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Resumo

A análise da geometria e da espessura da parede do ventrículo esquerdo é relevante para a caracterização da estrutura e função cardíacas. Contudo, a ressonância magnética cardíaca em eixo curto fornece uma representação bidimensional através de cortes espacialmente separados, tornando necessária a reconstrução tridimensional da geometria entre os cortes disponíveis. Embora a interpolação permita estabelecer continuidade entre contornos, não garante a preservação da anatomia ventricular. Além disso, a avaliação clínica da espessura miocárdica baseia-se frequentemente na avaliação de pontos específicos e na análise visual. Neste contexto, o modelo de 17 segmentos da American Heart Association (AHA) constitui uma referência para a avaliação regional do ventrículo esquerdo.

Nesta dissertação, propõe-se uma abordagem baseada em aprendizagem automática para a reconstrução tridimensional do ventrículo esquerdo a partir de contornos de ressonância magnética cardíaca. Utiliza-se uma representação neuronal implícita para descrever continuamente as superfícies endocárdica e epicárdica nas fases de telediástole e telessístole. O modelo é inicialmente treinado com formas ventriculares sintéticas e posteriormente ajustado com segmentações clínicas. A espessura da parede miocárdica é estimada a partir da geometria reconstruída, permitindo a análise local e regional segundo os 17 segmentos da AHA.

O desempenho da abordagem é avaliado por comparação com métodos baseados em funções de base radial e com um modelo estatístico de forma. Os resultados mostram que as superfícies reconstruídas acompanham os contornos observados, com maior concordância entre métodos na cavidade ventricular do que na parede miocárdica. Verifica-se também que as estimativas da espessura dependem da geometria reconstruída e do método de cálculo, pelo que uma correlação elevada não implica necessariamente concordância entre os valores obtidos. A análise segmentar permite ainda caracterizar as variações regionais da espessura entre a telediástole e a telessístole.

Este trabalho explora, assim, o potencial das representações neuronais implícitas na reconstrução tridimensional do ventrículo esquerdo e na quantificação da espessura miocárdica. Contudo, na ausência de uma referência anatómica tridimensional independente, não é possível determinar conclusivamente qual dos métodos produz a reconstrução mais próxima da anatomia real. Os resultados evidenciam, ainda assim, a influência das opções de modelação geométrica nas estimativas da espessura e a importância de considerar o método de reconstrução na interpretação dos resultados.

Palavras-chave: Aprendizagem automática em medicina; ressonância magnética cardíaca; reconstrução tridimensional; espessura miocárdica; representações neuronais implícitas; modelação cardíaca.

Abstract

Short-axis cardiac magnetic resonance imaging provides detailed observations of the left ventricle on spatially separated slices, leaving the geometry between them unknown. Reconstructing both ventricular boundaries is necessary to estimate myocardial wall thickness beyond the acquired planes, but interpolation alone does not guarantee anatomical accuracy.

This dissertation proposes a machine learning approach for three-dimensional left ventricular reconstruction from magnetic resonance contours. An implicit neural representation describes the endocardial and epicardial surfaces at end-diastole and end-systole. The model is trained on synthetic ventricular shapes and subsequently fine-tuned using surfaces fitted to clinical segmentations. The resulting geometry supports local thickness maps and regional analysis using the 17-segment model of the American Heart Association (AHA).

Comparisons with radial basis function reconstructions and a statistical shape model show greater agreement in the ventricular cavity than in the myocardial wall. Thickness estimates depend on both the reconstructed geometry and the measurement method, so high correlation does not necessarily imply agreement between values. Regional analysis also characterises changes in thickness between the two cardiac phases.

Without an independent three-dimensional anatomical reference, the evaluation cannot determine which approach reconstructs the true anatomy most accurately. The findings highlight the importance of considering reconstruction and measurement choices when interpreting myocardial wall thickness.

Keywords: Medical machine learning; cardiac magnetic resonance; three-dimensional reconstruction; myocardial wall thickness; implicit neural representations; cardiac modelling.

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List of Acronyms

Acronym	Definition
ACDC	Automated Cardiac Diagnosis Challenge
AHA	American Heart Association
CI	Confidence Interval
DSC	Dice Similarity Coefficient
ED	End-Diastole
EDT	Euclidean Distance Transform
ES	End-Systole
HCM	Hypertrophic Cardiomyopathy
HD95	95th-percentile Hausdorff Distance
ICC	Intraclass Correlation Coefficient
INR	Implicit Neural Representation
LoA	Limits of Agreement
LV	Left Ventricle
MAE	Mean Absolute Error
MRI	Magnetic Resonance Imaging
NIfTI	Neuroimaging Informatics Technology Initiative
NOR	Normal cohort
PCA	Principal Component Analysis
PDE	Partial Differential Equation
RAS+	Right-Anterior-Superior coordinate convention
RBF	Radial Basis Function
ReLU	Rectified Linear Unit
RMSE	Root-Mean-Square Error
RQ	Research Question
RV	Right Ventricle
SAX	Short-Axis
SD	Standard Deviation
SDF	Signed Distance Function
SSM	Statistical Shape Model

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CHAPTER 1

Introduction

This chapter motivates the reconstruction of continuous left-ventricular geometry from contours delineated on widely spaced cardiac magnetic resonance slices and defines the research problem. It then presents the research questions, the contributions of the thesis, and the organisation of the remaining chapters.

1.1. Motivation

Understanding cardiac structure is fundamentally a geometric problem. The left ventricle (LV), the chamber that pumps oxygenated blood into the body, is a three-dimensional structure whose shape, size, and wall thickness change continuously throughout the cardiac cycle. At end-diastole (ED), the ventricle has filled and is at its largest; at end-systole (ES), it has contracted, so the cavity is smallest and the muscular wall, the myocardium, is at its thickest. The difference between these two states is what quantifies ventricular pumping function: cavity volumes at ED and ES give the stroke volume and ejection fraction, while the volume of the myocardium gives its mass [1]. Beyond these global indices, the geometry of the ventricle carries diagnostic information of its own. Myocardial wall thickness is related to the diastolic behaviour of the ventricle [2], an abnormally thick wall is the defining feature of hypertrophic cardiomyopathy [3], and variations in ventricular shape and thickness have been associated with structural remodelling and cardiovascular risk at the population level [4, 5].

Cardiac magnetic resonance imaging (MRI) is the preferred imaging method for measuring these quantities. It is non-invasive, does not use ionising radiation, and offers good contrast between blood and myocardium, which is why it is regarded as the reference standard for ventricular volumes, mass, and function [1, 6]. In routine cine acquisitions, the LV is imaged in short-axis (SAX) planes perpendicular to its long axis, stacked from the base, the wide end where the valves sit, towards the apex, the pointed tip, and repeated over the cardiac cycle so that both ED and ES are captured.

The images themselves, however, do not yield clinical measurements directly. Volumes, mass, and wall thickness are computed only after the relevant boundaries have been outlined on every slice: the endocardium, the inner boundary of the blood-filled cavity, and the epicardium, the outer boundary of the myocardium. This outlining, called segmentation, is therefore the step on which all later measurements depend. Drawing these outlines manually is time-consuming,

and the outlines can differ between experts or when the same expert repeats the task [6]. These differences limit reproducibility, while the time required makes manual segmentation less practical as the number of examinations grows. These limitations have motivated a sustained effort towards automated segmentation, driven by public benchmarks and challenges [1, 7, 8]. The need is most evident in large cohorts such as the UK Biobank, where thousands of subjects must be analysed consistently [9], and where fully convolutional networks now produce segmentations whose derived measurements approach the agreement observed between human experts [10].

Segmentation, however, answers a two-dimensional question. It produces a set of contours on separate planes, whereas the quantities of clinical interest describe a continuous three-dimensional organ. Regional analysis, such as reporting wall thickness according to the American Heart Association (AHA) 17-segment model [11], and the construction of population atlases that compare anatomy across subjects [5, 12] both require a description of the ventricle that extends beyond the individual slices. How that description is obtained from segmented contours is the problem addressed in this thesis.

1.2. Problem Statement

Reconstructing the LV in three dimensions from segmented SAX contours is a long-standing problem in cardiac image analysis. A cine SAX stack typically resolves the ventricle at 1–2 mm within each slice, but consecutive slices lie 6–10 mm apart [6]. These contours are sparse: they describe the ventricle in detail within each slice but leave the geometry between slices, including the curvature of the apex and the closure of the base, unobserved. Many surfaces can pass through the same contours, so the reconstruction depends on the assumptions used to fill the gaps.

Several families of methods supply these assumptions. Contour lofting joins corresponding points on adjacent slices and passes exactly through the observations, but produces piecewise-linear geometry between them [6]. Statistical shape models (SSMs) restrict the surface to the modes of variation of a training population [13, 14, 15], which stabilises sparse reconstructions but can draw atypical anatomy towards the population mean. Radial basis function (RBF) interpolation defines the surface as the zero level set of a smooth function fitted to the contour points [16], trading regional detail for smoothness. Implicit neural representations (INRs) instead encode shapes as zero level sets of functions computed by neural networks [17, 18] and have been used to reconstruct cardiac surfaces from widely spaced images and sparse segmentations [19, 20]. Closed LV surfaces from sparse contours are therefore not new; the approaches differ in the prior that fills the unobserved geometry and in how far measurements reflect the data rather than that prior.

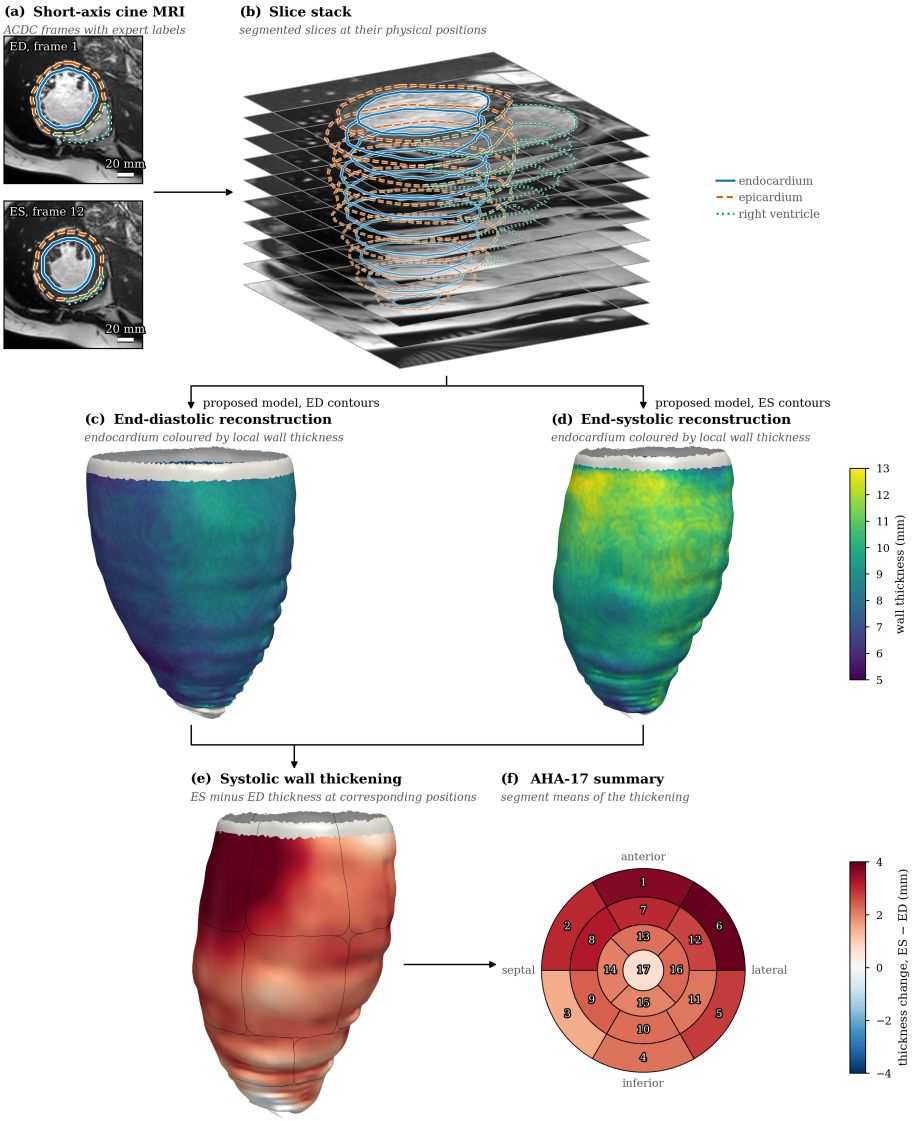

Figure 1.1. Overview of the proposed pipeline. (a) Segmented SAX slices at ED and ES. (b) The slices placed along the LV long axis form a sparse stack of contours, from which the proposed model reconstructs endocardial and epicardial surfaces for (c) ED and (d) ES, coloured by local wall thickness. (e) Wall thickening from ED to ES on the surface, with AHA 17-segment boundaries, and (f) its summary per segment.

The difference matters most for myocardial wall thickness. The endocardium and epicardium must form a consistent pair whose separation remains positive and spatially coherent; independently reconstructed surfaces may be plausible yet intersect or collapse the wall between them [19]. Priors are also often tied to a single cardiac phase, so a shape model built from diastolic anatomy imposes a diastolic shape space on systolic observations. A discrepancy of one or two millimetres between reconstructions is small relative to the cavity but substantial relative to the wall. Surface-distance and overlap metrics, the usual criteria for reconstruction quality [21], do not by themselves show how these discrepancies propagate into thickness measurements.

Public data impose a further constraint. Benchmarks such as the Automated Cardiac Diagnosis Challenge (ACDC) and the second Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Image Segmentation Challenge (M&Ms-2) provide images and segmentation masks for individual slices rather than complete three-dimensional LV meshes [1, 22]. Without an independent three-dimensional reference, a learned model must be supervised with surfaces fitted to the same contours and assessed through fidelity to the observed slices and agreement with alternative reconstructions.

This thesis addresses the problem with a phase-conditioned INR, trained on synthetic LV geometries sampled from an SSM and on meshes derived from real cardiac MRI segmentations, as illustrated in Figure 1.1. It adapts the implicit formulation rather than introducing a new representation, taking sparse contours as input and representing both surfaces as continuous three-dimensional functions that can be queried beyond the acquired slices. Conditioning on cardiac phase lets one model represent both ED and ES, while defining the epicardium through a positive offset from the endocardium keeps the outer surface outside the inner one by construction.

Because no independent reference exists, the evaluation is comparative. The reconstruction is assessed against the observed contours and against RBF and SSM reconstructions of the same contours, with slices removed from the input to measure sensitivity to slice removal. Wall thickness is measured on each geometry with several methods, separating the effect of the reconstruction prior from that of the measurement method. The aim is not to replace clinical interpretation, but to investigate whether the resulting geometry preserves quantities relevant to cardiac analysis and provides a more consistent basis for quantitative analysis of sparse cardiac MRI.

1.3. Research Questions

As Section 1.2 shows, a closed LV surface can already be obtained from sparse contours by RBF interpolation, SSM fitting, or an implicit neural model. The open questions concern what a learned prior contributes relative to these alternatives, whether it can satisfy the paired-surface and multi-phase requirements of wall-thickness analysis, and how strongly the resulting measurements depend on choices made during reconstruction and measurement. This thesis therefore addresses the following research questions (RQs):

RQ1. How does an implicit reconstruction learned from synthetic and clinical LV geometry compare with RBF and SSM reconstructions of the same sparse SAX contours in fidelity to the observed contours, agreement on the geometry inferred between slices, and sensitivity to the number of available slices?

RQ2. Can a single phase-conditioned model reconstruct paired endocardial and epicardial surfaces at both ED and ES while maintaining a positive wall thickness, without fitting a separate model to each phase or reconstructing the two surfaces independently?

RQ3. To what extent does myocardial wall thickness measured on the reconstructed geometry depend on the reconstruction prior and on the thickness method, and are regional and pathology-related differences in thickness retained?

1.4. Contributions

The thesis makes three contributions: a shared process for preparing training data, an adaptation of implicit reconstruction to paired LV surfaces, and an evaluation approach that distinguishes geometric and measurement consistency.

A shared data pipeline is developed for synthetic LV shapes sampled from an SSM and clinical ED and ES segmentations from ACDC and M&Ms-2. Both sources are converted into input contours and target surfaces in the same format, allowing synthetic pre-training and clinical fine-tuning without changing how the data are represented. The clinical surfaces are fitted to the observed contours and used to train the proposed model; they are not independent three-dimensional anatomical ground truth.

The proposed model adapts established point-set and implicit-representation methods by summarising endocardial and epicardial contour evidence separately and combining a global shape description with local contour information. A cardiac-phase input allows one model to represent both ED and ES, while a positive offset links the endocardial and epicardial fields. This offset is trained directly using the separation between the target surfaces. The contribution is this task-specific combination, not the underlying methods themselves.

An evaluation approach is also established to separate reconstruction from measurement. It examines fidelity to the observed contours, agreement with RBF and SSM reconstructions of the same inputs, and sensitivity to slice removal. Four wall-thickness methods are compared on the same geometry before local measurements are summarised in AHA-17 regions and examined across cardiac phases and patient groups. In the absence of independent three-dimensional ground truth, these comparisons characterise consistency and dependence on methodological choices rather than establish anatomical accuracy or diagnostic performance.

1.5. Thesis Organisation

The remaining chapters follow the progression from the reconstruction problem to its evaluation. Chapter 2 reviews cardiac MRI-based reconstruction and myocardial wall-thickness measurement, providing the context for the methodological choices. Chapter 3 describes the data preparation, the proposed model, its training and inference, and the methods used to measure wall

thickness. Chapter 4 evaluates the reconstructed geometry, how much it changes when slices are removed, and the resulting local and regional thickness measurements. Finally, Chapter 5 answers the research questions, summarises the contributions and limitations, and identifies directions for future work.

CHAPTER 2

Literature Review

This chapter describes the review methodology and examines prior work on reconstructing paired LV surfaces from sparse cardiac MRI observations and measuring the myocardial wall represented by those surfaces.

2.1. Review Methodology

The review was designed to compare methods relevant to sparse cardiac MRI, three-dimensional shape reconstruction, geometric learning, and myocardial wall-thickness measurement, rather than to provide an exhaustive systematic review.

Scopus, PubMed, and IEEE Xplore were searched, with Google Scholar used for backward and forward citation chaining. The search included foundational work published from 2000 onwards and selected studies available up to 2026. Although recent work was prioritised, earlier publications were retained where they establish concepts used by later methods. The resulting review is selective and does not claim exhaustive coverage of the literature published through 2026.

Search terms combined anatomy and imaging concepts (“cardiac MRI”, “left ventricle”, “short-axis”, “SAX”, “myocardium”) with reconstruction terms (“3D reconstruction”, “mesh reconstruction”, “shape model”, “implicit surface”, “signed distance function”), learning terms (“deep learning”, “graph neural network”, “point cloud”, “neural representation”), and measurement terms (“wall thickness”, “myocardial thickness”, “Laplace equation”, “transmural”).

The refined query applied consistently across databases was:
(“cardiac MRI” OR “left ventricle” OR “short-axis”) AND (“3D reconstruction” OR “statistical shape model” OR “signed distance function” OR “graph neural network” OR “implicit representation”) AND (“wall thickness” OR “mesh” OR “deep learning”)

Studies were included when they addressed LV three-dimensional reconstruction or shape modelling, proposed or evaluated myocardial wall-thickness methods, developed geometric learning methods applicable to anatomical meshes, or introduced benchmark cardiac MRI data relevant to LV analysis. Studies focused only on segmentation, on non-cardiac anatomy without transferable methodology, or lacking peer review or sufficient methodological detail were excluded. Publications before 2000 were considered only when they provided foundational methods directly relevant to the reviewed approaches.

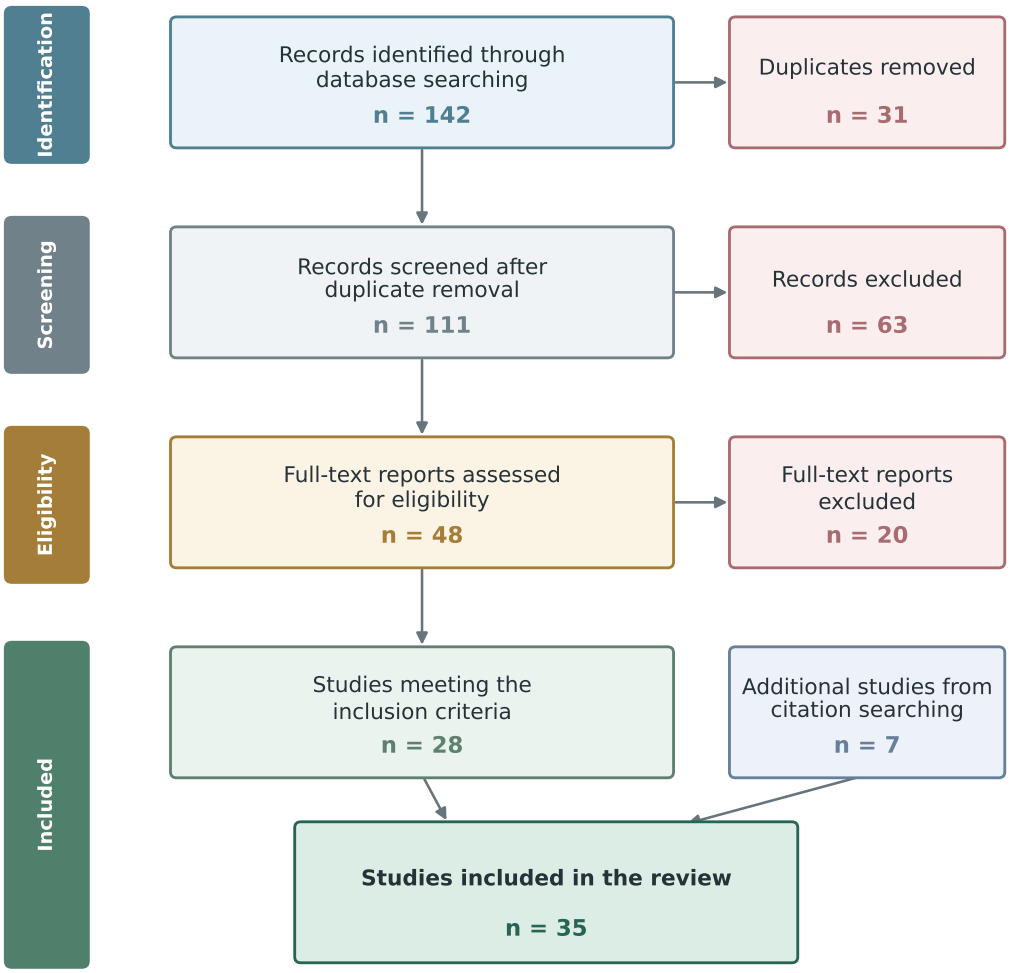

Figure 2.1. PRISMA flow diagram summarising the selection of the 35 publications included in the review.

The search returned 142 records. After duplicate removal and screening, 28 studies met the selection criteria; backward citation searching added 7, giving a final corpus of 35 publications. Figure 2.1 summarises this process as a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram [23].

The selected studies cover cardiac MRI data and segmentation, SSMs, learned 3D reconstruction, and wall-thickness measurement. The next section relates these areas to the specific problem of recovering paired LV surfaces from sparse contours.

2.2. Technical Background

The central distinction in this review is between anatomy observed on acquired MRI slices and geometry that must be inferred between them. This distinction connects the constraints of paired LV surface reconstruction to the available representations of three-dimensional anatomy.

Cine cardiac MRI captures the LV at several points in the cardiac cycle. Of these, ED, when the cavity is largest, and ES, when it is smallest, provide a useful description of ventricular contraction and are the phases labelled in benchmarks such as ACDC [1].

In a SAX acquisition, each plane samples a cross-section perpendicular to the ventricular long axis. The anatomy is resolved in detail within a slice, but the slices are much farther apart than the pixel spacing within each slice, producing an unevenly sampled image stack [6]. Segmentation can therefore recover the myocardium where it is observed without determining how its boundaries curve, taper, or change in the unobserved intervals.

Each slice contributes an endocardial contour around the cavity and an epicardial contour around the outer myocardium. In three dimensions, these must form nested surfaces with a positive separation. A reconstruction that treats the surfaces independently may produce individually plausible boundaries while still creating an intersection, reversal, or local collapse in the wall between them, a difficulty also addressed by recent joint implicit reconstructions of cardiac structures [19].

The contours alone do not determine one unique continuous surface: many smooth surfaces can pass through the same contour rings. Smoothing stabilises the reconstruction but can remove genuine regional variation; a strong population prior can bridge missing regions but may pull abnormal anatomy towards common shape; and close contour fitting can preserve segmentation noise. A useful method must balance these competing constraints rather than optimise any one of them in isolation [6, 15].

The same reasoning determines how reconstruction should be evaluated. The surfaces should agree with observed anatomy, remain closed where volume is needed, preserve subject-specific shape, and form a valid wall at both phases. No single overlap or surface-distance measure assesses all of these properties, so geometric distance, volumetric agreement, topology, and the stability of wall-derived measurements must be considered together [15, 19].

The representation chosen for three-dimensional anatomy determines both what can be measured and how missing geometry is inferred. Voxel volumes are well suited to image-based segmentation and overlap analysis, but retain the spatial limits of the sampling grid. Surface meshes support direct geometric and regional measurements, although remeshing, smoothing, and repair can alter the anatomy being evaluated. Neither representation alone determines the geometry between sparsely acquired slices.

SSMs address this missing information by constraining meshes with population-level modes of variation [13, 15]. They can therefore provide a geometric prior or synthetic training examples, but remain limited by the topology and anatomical variation represented in the source atlas.

Implicit methods offer a different compromise. DeepSDF and Occupancy Networks represent geometry as continuous implicit fields [17, 18], from which meshes can be extracted using

methods such as Marching Cubes [24]. Work on Eikonal regularisation and Fourier features further addresses distance-field consistency and the recovery of local spatial variation [25, 26]. By improving field consistency and spatial detail, these developments make implicit representations relevant to sparse-contour reconstruction. The extracted surface nevertheless remains dependent on the learned field and extraction procedure.

Paired LV surfaces introduce an additional constraint: two independently predicted fields do not guarantee a valid ordering. Defining one surface in relation to the other through a positive separation instead represents a reference boundary and a constrained wall. Joint implicit reconstruction of cardiac structures provides one way of modelling related boundaries together, although surface ordering and wall validity must still be assessed explicitly [19].

2.3. Related Work

This section compares how the reviewed methods address the evidence available on acquired slices, the geometry missing between them, and the measurements derived from the reconstructed wall.

2.3.1. Clinical observations and conventional geometric priors

SAX cine MRI is the standard protocol for LV evaluation, producing 2D slices perpendicular to the long axis with high in-plane resolution (1–2 mm) but large inter-slice gaps (6–10 mm) [6]. The resulting anisotropy creates the central reconstruction problem: the observed contours are detailed within each slice, but the geometry between slices, especially near the apex and base, must be inferred. The AHA 17-segment model provides a common anatomical frame for reporting regional measurements [11].

Modern segmentation networks can identify endocardial and epicardial boundaries reliably on the acquired slices. U-Net established the encoder–decoder pattern used by many medical segmentation systems, while nnU-Net showed that much of their configuration can be adapted automatically to a dataset [27, 28].

Although both methods improve the observed contours, they still optimise agreement on sampled image planes. High slice-wise overlap can therefore coexist with uncertain through-plane geometry. Moreover, because the contours become constraints on the reconstructed surface, segmentation errors can propagate into regions where no image evidence is available rather than remaining local to one slice.

ACDC provides expert ED and ES labels across five diagnostic groups, including normal ventricles and hypertrophic cardiomyopathy (HCM) [1]. Its phase and diagnostic variation make it useful for testing whether a reconstruction retains contraction-related and abnormal shape differences. M&Ms-2 focuses on right-ventricular segmentation and extends evaluation across centres, scanner vendors, and diseases [22]. For reconstruction, its multi-source design shows

that performance on one acquisition source does not guarantee transfer to another. A plausible surface on one cohort is insufficient evidence of generalisation because acquisition variation affects the contours from which the entire surface is inferred.

At population scale, the UK Biobank provides a large and standardised cardiac MRI resource [9]. Automated segmentation of this cohort has enabled large anatomical studies and statistical atlases [10]. Together, slice-wise clinical observations and population-level atlases provide complementary forms of anatomical evidence.

The two forms are not interchangeable. Clinical labels retain subject-specific and pathological variation, but their through-plane geometry is only indirectly observed. A population atlas supplies complete and consistent surfaces, but it reflects the cohort and processing choices from which it was built. Combining them can reduce the weakness of either source alone, provided that the atlas is used as a prior rather than treated as patient-specific ground truth.

SSMs encode anatomical variability by matching the same anatomical locations across the training population and applying principal component analysis (PCA) to extract the dominant modes of variation [13, 15]. They provide a compact description of plausible shape, but their quality depends on how accurately the same anatomical locations are matched, on alignment, and on the diversity of the population used to construct them.

Cardiac SSMs have progressed from multi-object ventricular models to large population atlases. Matching the same points across subjects allows related structures to be represented consistently [14], while the UK Digital Heart Project atlas captures population variation in ventricular shape and motion [12]. Their modelling of coordinated variation across related cardiac structures is directly relevant to paired-surface reconstruction, where the two boundaries cannot be treated in isolation.

Shape-model studies have also identified variation associated with hypertrophic remodelling, normal anatomical diversity, and myocardial infarction [4, 5, 29]. Their findings show that clinically relevant anatomy is not captured by global volume alone. A population prior must therefore retain regional differences in proportion, apical form, and curvature rather than draw every case towards one mean geometry.

SSMs have two principal roles. They can constrain reconstruction to a learned shape space, or they can generate synthetic geometries by sampling that space [15, 30]. Synthetic sampling provides complete, plausible meshes that can increase the amount of geometric supervision available for learning-based methods.

Synthetic sampling increases the amount of complete geometric supervision, but it cannot introduce anatomy absent from the source atlas. It may also under-represent rare or severe pathology when the retained PCA modes favour common variation. Clinical data remain important for adapting a population prior to subject-specific and pathological anatomy while retaining

the surface continuity learned from complete meshes. Combining SSM-derived and clinical data therefore assigns complementary roles to population-level priors and observed anatomical variation.

Before a learned model can be supervised with clinical data, the segmented SAX contours must be related to a three-dimensional surface. Linear contour interpolation, SSM fitting, and RBF fitting offer three different ways to supply the missing through-plane geometry. Their principal difference is the source of the reconstruction prior: linear interpolation uses only neighbouring contours, an SSM uses population-level modes of anatomical variation, and an RBF fit imposes smoothness through a continuous implicit field.

Linear contour lofting first matches points between adjacent rings and then joins them with triangular strips. It is transparent, computationally inexpensive, and passes directly through the observed contours. However, the geometry between two slices is piecewise linear and depends on how the starting points are aligned and which points are matched between rings. Large inter-slice gaps therefore produce abrupt changes in slope, while the apex and base require explicit closure rules. These limitations make lofting useful as a direct interpolation baseline, but less suitable as the sole source of smooth three-dimensional supervision from anisotropic SAX data [6].

SSM fitting instead aligns a template with the observations and optimises its shape coefficients so that the deformed surface approaches the contours. The result retains a fixed topology and population-informed anatomical plausibility, which are valuable when observations are sparse [13, 15]. The same prior also limits the fit: anatomy outside the variation represented by the atlas cannot be recovered faithfully, and fitting requires a model appropriate to the structure and cardiac phase under study. An ED-only SSM cannot provide equivalent supervision for ES without imposing a diastolic shape space on systolic observations.

RBF reconstruction represents the surface as the zero level set of a smooth function fitted to scattered points [16]. Surface samples are assigned zero values, while additional samples displaced to either side of the contour provide the sign needed to distinguish interior from exterior. The resulting implicit field can be evaluated on a regular grid. Marching Cubes then extracts a mesh from the sampled field [24].

Unlike SSM fitting, this procedure does not require a phase-specific template or fixed mesh correspondence. Unlike linear lofting, it defines the intervening geometry as a smooth field rather than as strips between paired vertices. Its limitations are different: the reconstruction depends on the kernel, smoothing, and off-surface constraints, and strong regularisation can suppress genuine local variation.

2.3.2. Deep learning models for 3D shape reconstruction

Classical reconstruction methods depend on the sampled contours and on the assumptions imposed between slices, particularly near the apex and base, as discussed in Section 2.3.1. Learned reconstruction methods address the same trade-off between observation fidelity and geometric prior through two main families: template deformation and implicit field prediction.

Template methods learn to displace the vertices of a fixed mesh. Pixel2Mesh and Pose2Mesh demonstrate that graph-based deformation can recover dense geometry from sparse image or landmark evidence [31, 32]. In cardiac applications, fixed correspondence is valuable for population analysis and regional measurement [7, 12, 33]. The trade-off is that the output remains tied to the topology and resolution of the chosen template. The same constraint that simplifies correspondence can therefore limit how far the reconstruction departs from the anatomy encoded by that template.

INRs remove the fixed output template by learning a continuous field from which a surface can be extracted at a chosen resolution. DeepSDF and Occupancy Networks established this approach using signed-distance and occupancy formulations, respectively [17, 18]. For sparse cardiac reconstruction, removing the predefined mesh avoids restricting the output geometry to fixed resolution and connectivity. Output flexibility does not, however, guarantee that two predicted LV surfaces remain correctly ordered; anatomical validity must be addressed separately.

Recent cardiac work shows why implicit representations are useful when MRI slices are far apart. In a SAX scan, the detailed shape of the ventricle is observed within each slice, but less information is available between slices. Brás et al. [19] addressed this anisotropy problem with a neural implicit representation that reconstructs the LV and the right ventricle (RV), including the endocardial and epicardial boundaries of the LV myocardium. The study is closely related to sparse-contour reconstruction because both approaches seek continuous cardiac shape from incomplete observations. Brás et al. use image data directly, whereas contour-based approaches operate on segmented point observations.

Muffoletto et al. [20] also reconstruct three-dimensional cardiac shape from sparse segmentations with a neural implicit representation. Their method uses a learned biventricular shape prior and includes both SAX and long-axis views. It is therefore broader than LV-focused signed distance function (SDF) reconstruction. At the time of writing, the authors provide a public project repository, although a complete code and dataset release could not be verified.

The INR literature also provides mechanisms relevant to thin cardiac anatomy. Fourier features improve the recovery of local spatial variation, while Eikonal regularisation promotes a consistent distance field [25, 26]. In sparse-slice settings, both mechanisms can help mitigate the loss of local surface detail associated with interpolation across large gaps.

Recent cardiac INRs also model motion across the cardiac and respiratory cycles [34, 35, 36, 37]. Their main goal is temporally coherent image or volume reconstruction, whereas other implicit approaches focus specifically on continuous anatomical surface reconstruction conditioned on cardiac phase.

PointNet provides a complementary way to encode unordered observations [38]. Its predictions do not change when the input points are listed in a different order, which is useful for SAX contours because their point ordering is arbitrary.

2.3.3. Wall-thickness measurement techniques

Myocardial wall thickness is a primary clinical biomarker for cardiac remodelling. Grossman et al. [2] demonstrated that wall thickness is a principal determinant of left-ventricular diastolic stiffness, with hypertrophied ventricles exhibiting markedly elevated stiffness proportional to ED wall thickness. More recently, Huelnhagen et al. [39] showed at 7.0T MRI that wall thickness correlates with tissue transverse relaxation properties, providing a non-invasive way to assess myocardial tissue properties. The AHA 17-segment model [11] provides the standard framework for reporting regional thickness values.

The accuracy of wall-thickness measurement depends critically on the quality of the underlying surface reconstruction, as any smoothing or geometric artefact propagates directly into the thickness estimate. The literature describes five families of measurement methods, each with distinct trade-offs.

The simplest approach uses nearest-neighbour distance: for each endocardial vertex, a k-d tree [40] identifies the closest epicardial point, yielding a measurement in $O(N \log M)$ time. Its runtime depends on mesh size and the implementation. Nearest-neighbour distance can provide a lower bound on the true thickness across the wall, also called transmural thickness, because it ignores the anatomical direction of measurement — the nearest point may lie tangentially rather than across the wall.

Ray-based methods address this limitation by casting a ray from each endocardial vertex along its surface normal, the direction perpendicular to the surface, until it intersects the epicardium. The intersection is computed via the Möller–Trumbore algorithm [41], accelerated with a bounding volume hierarchy to $O(\log F)$ per query. Measuring along the surface normal better approximates the direction across the wall but remains sensitive to errors in the normal: a slight rotation can make the measured distance too long or too short. The shape diameter function variant [42] reduces this sensitivity by casting a cone of K rays (typically $K = 7$) at a half-angle of 30° around the normal and reporting the median intersection distance, effectively filtering outlier hits caused by local mesh irregularities.

Volumetric methods operate on voxel grids. The Euclidean Distance Transform (EDT) computes, for each myocardial voxel, the distance to the nearest boundary, and the sum of the endocardial and epicardial distance fields approximates local thickness ($t = D_{\text{endo}} + D_{\text{epi}}$). The estimate is exact for planar parallel boundaries but can be biased in curved regions because the nearest-boundary directions do not necessarily follow a common path across the wall. The method is computationally efficient ($O(V)$ via the Saito–Toriwaki algorithm [43]) but limited by voxel resolution.

Partial differential equation (PDE)-based methods are widely used to estimate thickness across tissue layers. In a method developed for cortical rather than cardiac anatomy, Jones et al. [44] solved Laplace's equation between the two tissue boundaries and used the resulting non-intersecting field lines to establish corresponding paths. The same geometric principle can be transferred to myocardial boundaries, with thickness obtained from the path length between the boundaries along these streamlines. Yezzi and Prince [45] developed a PDE method that calculates distance fields spreading from both boundaries along the Laplacian field. Adding these distances gives the local tissue thickness.

Finally, mesh-based methods that match points between the surfaces and smooth the resulting measurements combine an initial geometric estimate (normal projection or nearest neighbour) with iterative Laplacian smoothing on the mesh connectivity graph. At each iteration, each vertex's thickness is updated as a weighted average of its neighbours, anchored to the initial estimate. Regularisation produces spatially smooth thickness maps suitable for bullseye visualisation, but too much smoothing can reduce genuine local differences in thickness.

Wall-thickness values depend on both the measurement method and the reconstructed surfaces. Smoothing may make a measurement more stable, but it can also change the distance between the endocardial and epicardial surfaces. Similarly, an irregular surface can create thickness variation that does not reflect the myocardium. For this reason, thickness methods should be compared on the same surfaces. Reconstructions should also be compared using the same thickness method. The present work evaluates both reconstruction consistency and agreement between thickness methods.

2.3.4. Summary and research gap

The reviewed literature shows that each component of the LV analysis pipeline is well-studied in isolation: deep learning gives accurate slice-wise segmentation, SSMs encode population-level shape priors, implicit representations provide continuous surface modelling, and multiple wall-thickness algorithms exist. However, limited work integrates these components into a single pipeline for sparse SAX contour-based reconstruction.

A gap remains in methods that jointly address sparse SAX contour encoding, continuous paired endocardial and epicardial surface reconstruction, cardiac-phase conditioning, preservation of positive wall separation, and training with both clinical observations and SSM-derived synthetic geometries. Such an integrated framework would combine the complementary strengths of population-based shape priors, continuous implicit representations, and clinical contour data while supporting wall-thickness measurement beyond the acquired slices.

There is also an evaluation gap. When no complete three-dimensional reference is available, contour agreement only tests the acquired slices. It cannot validate the reconstructed shape between slices. Evaluation should therefore distinguish contour fidelity from agreement between reconstruction methods. Thickness values must also be interpreted as measurements on an inferred wall, not as direct observations. These principles guide the evaluation in the following chapters.

CHAPTER 3

Methodology

This chapter describes how the proposed machine learning model reconstructs the LV in 3D from contours drawn on a few SAX slices separated by gaps, and how myocardial wall thickness is then measured on the reconstructed surfaces. The proposed reconstruction model learns first from synthetic hearts and then from clinical scans, and several established thickness methods are compared before regional results are reported.

3.1. Overview

Wall thickness is measured between the inner and outer LV surfaces, the endocardium and epicardium. SAX images show these boundaries only on separate slices [1, 46], leaving the shape between them unknown. Reconstructing continuous surfaces allows thickness to be examined across the ventricle rather than only at the acquired slices. Figure 3.1 shows how the proposed model learns to fill these gaps, recover the inner and outer surfaces, and provide the geometry for wall-thickness measurement.

Training follows the stages at the top of the figure. It begins with synthetic ED hearts generated by an SSM, which describes how ventricular shape varies across a population [12, 13]. These hearts are sliced into contours, giving the model examples with known complete surfaces (a). Because the full shape is available, the model can learn how the visible contours relate to the parts of the wall between them.

The knowledge gained from synthetic hearts is then carried into training on clinical MRI contours at ED and ES (b). Smooth surfaces fitted to those contours provide the targets for fine-tuning, introducing patient differences and the change in shape during contraction. The clinical targets are useful for learning, but they are not an independent reference for the anatomy between slices, since they are built from the same incomplete observations.

In panel (c), an encoder summarises the input contours, and an implicit decoder uses that description to represent the boundaries throughout space [17]. The inner and outer surfaces are learned together, with cardiac phase supplied as input so that the same model can reconstruct both ED and ES. The contours guide the reconstruction, while the shapes seen during training help determine the unobserved parts of the wall.

The bottom panels follow a patient from observed contours (d) to reconstructed surfaces (e) and local wall-thickness maps (f). Reconstruction and measurement remain separate. Different

ways of measuring across a curved wall can give different values, so thickness methods are compared on the same surfaces.

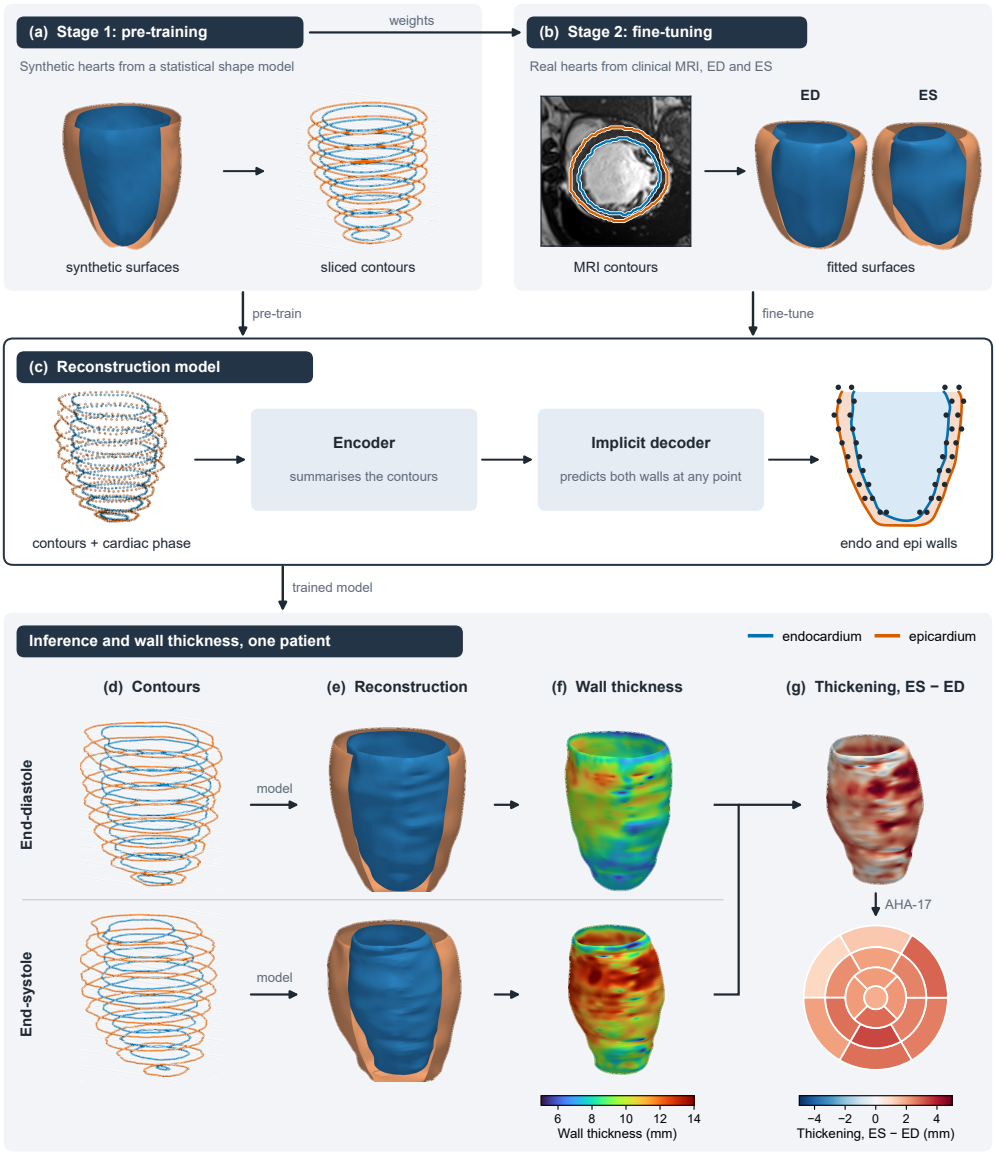

Figure 3.1. Overview of the methodology. (a) Pre-training on synthetic ED surfaces and their SAX contours. (b) Fine-tuning on clinical ED and ES contours with fitted target surfaces. (c) Phase-conditioned reconstruction of the endocardial and epicardial surfaces. (d)–(f) Clinical inference from observed contours to reconstructed surfaces and local Laplace wall thickness at ED and ES, which share a colour scale. (g) Systolic thickening (ES minus ED) on the ES endocardium and the corresponding difference in AHA segment means. Surfaces are smoothed for display only.

The maps shown here use Laplace thickness, with a shared ED and ES colour scale that makes changes between phases visible. Panel (g) shows systolic thickening, ES minus ED, while the bullseye summarises these changes across the AHA regions. These outputs allow local and regional patterns to be examined, but do not by themselves establish the accuracy of the reconstruction or the measurements.

The following sections describe data preparation (Section 3.2), the model architecture and its adaptations (Section 3.3 and Table 3.3), training and surface extraction (Section 3.4), and thickness measurement (Section 3.5). The outputs are evaluated in Chapter 4 to address the research questions in Section 1.3.

3.2. Data Preparation

Learning to fill the gaps between slices requires examples in which the observed contours are paired with the closed 3D surfaces they came from. The two branches in Figure 3.1 (a) and (b) obtain such pairs from opposite ends: the synthetic branch starts from complete shapes and slices them, whereas the clinical branch starts from slices and builds surfaces from them. ACDC and M&Ms-2 supply images and slice-wise segmentation masks, label images in which every pixel is marked as cavity, myocardium, or background, but not the closed ("watertight") endocardial and epicardial surfaces that a signed-distance decoder needs as supervision. The UK Digital Heart Project LV SSM[1] with clinical ED and ES segmentations from ACDC and M&Ms-2. The synthetic branch in Figure 3.1 (a) provides 1300 paired endocardial and epicardial surfaces spanning a controlled range of healthy shapes for pre-training. In the clinical branch (b), smooth implicit surfaces are fitted to the segmented contours to provide patient-specific targets for fine-tuning.

3.2.1. Synthetic ED geometry from the statistical shape model

The synthetic stream draws on the UK Digital Heart Project LV SSM of Bai et al. [12], built from more than 1,000 high-resolution cardiac MRI scans of healthy subjects. The statistical shape model describes the ED phase only. Its mean mesh contains two closed surfaces, the endocardium and the epicardium, with 22 043 vertices in total (the corner points of the triangles that make up the mesh). The mesh is accompanied by 100 principal modes of variation and their 100 eigenvalues λ_i, each measuring how much of the population's variation its mode explains. A new ED mesh, containing both surfaces, is generated by

$$X(b) = \bar{X} + \Phi b, \tag{3.1}$$

where $\bar{X}$ is the mean shape flattened to a vector, Φ is the matrix of mode vectors, and $b = [b_1, \ldots, b_{100}]^\top$ is the sampled coefficient vector. The standard deviation (SD) $\sigma_i = \sqrt{\lambda_i}$ sets the scale of variation along mode i. The first modes capture the largest sources of variation in the population: mode 1 mainly controls overall LV size, mode 2 the elongation along the long axis, and higher modes increasingly local deformations (Figure 3.2).

[1] https://github.com/UK-Digital-Heart-Project/Statistical-Shape-Model

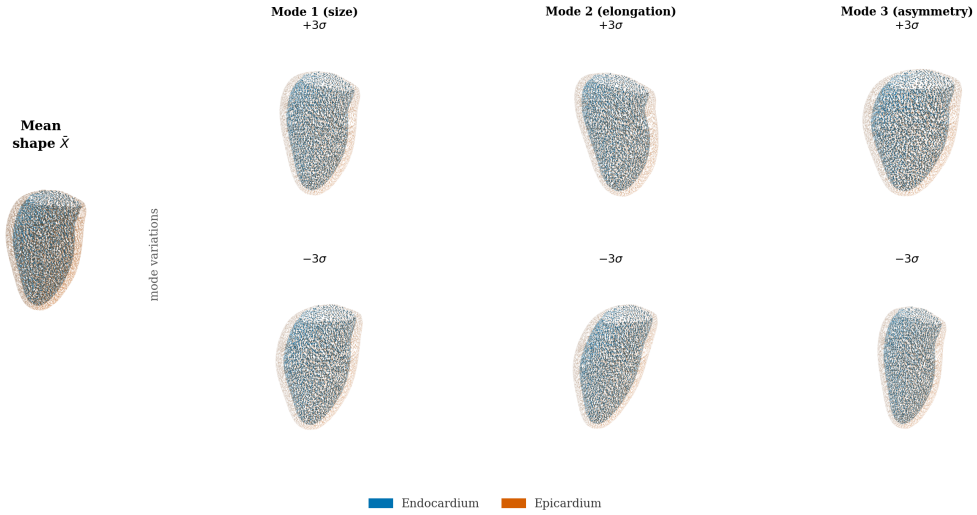

Figure 3.2. UK Digital Heart Project LV SSM mean shape and first three PCA modes at $\pm 3\sigma$, with the endocardium in blue and the epicardium in orange.

Of the 100 modes, the first 32 capture 99% of the shape variance and define the sampling space. Each candidate coefficient vector b was drawn by sampling each active component as $b_i \sim \mathcal{N}(0, \sigma_i^2)$ and was accepted only if it satisfied

$$\left|\frac{b_i}{\sigma_i}\right| \leq 3, \qquad \sum_i \left(\frac{b_i}{\sigma_i}\right)^2 \leq \chi^2_{0.99,d}, \tag{3.2}$$

where d is the number of active modes and $\chi^2_{0.99,d}$ is the 99th percentile of the chi-squared distribution with d degrees of freedom. The first condition keeps every mode within three SDs of the mean. The second limits the overall Mahalanobis distance [47], which measures the combined deviation after each mode has been scaled by its population variability. The threshold retains combinations within the 99th percentile of the SSM distribution.

Statistical plausibility does not guarantee anatomical plausibility, so each accepted vector is turned into a mesh with Equation (3.1) and checked once more. Meshes whose volume, surface area, or roundness (sphericity) fall outside limits derived from the population, or whose apex and base point the wrong way, are discarded; panel (f) of Figure 3.3 contrasts an accepted shape with a rejected outlier. Of 2048 candidates, 1300 (63%) passed both the statistical and the geometric checks.

Because both surfaces belong to the same mesh, a single coefficient vector deforms the cavity and the outer wall together, and every accepted sample already forms an ED endocardial and epicardial pair. On the mean shape, the distance from each endocardial vertex to the epicardium has a median of 6.1 mm, as shown by the histogram in panel (d) of Figure 3.3. These pairs give the decoder a consistent inner and outer surface during pre-training, so that the proposed reconstruction model learns the general relation between the two LV boundaries before

it sees clinical data. As the SSM covers only end-diastole, the synthetic stream contributes no systolic geometry, and the synthetic wall is never used as a reference for any thickness result.

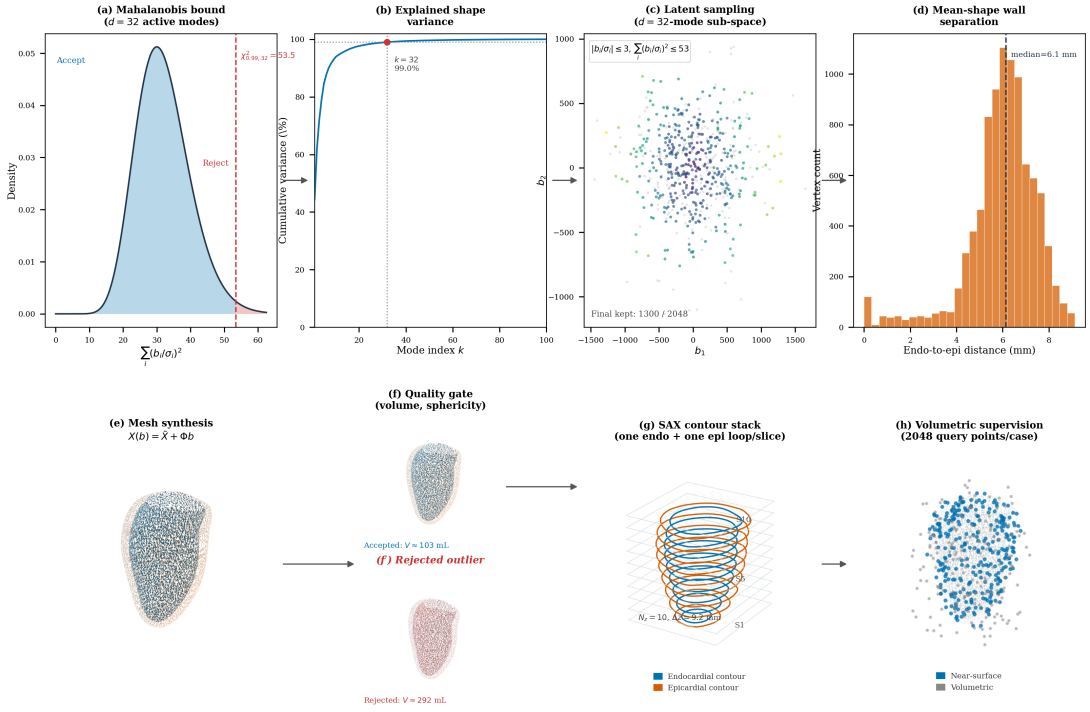

Figure 3.3. Synthetic ED dataset generation, from SSM sampling to geometric supervision. Top row, statistical constraints: (a) distribution of the overall-deviation (Mahalanobis) statistic of Equation (3.2) with the $\chi^2_{0.99,32}$ acceptance threshold; (b) cumulative shape variance, with 32 active modes reaching 99%; (c) accepted coefficient samples in the first two modes; (d) distribution of the endocardial-to-epicardial distance on the SSM mean shape. Bottom row, geometric construction: (e) mesh synthesis via Equation (3.1); (f) an accepted shape next to a rejected outlier; (g) the 10 SAX contour levels; (h) the 2048 near-surface and volumetric query points used for decoder supervision.

Finally, each surface pair was converted into the input and supervision required by the proposed reconstruction model. Slicing both surfaces at 10 evenly spaced levels along the long axis produces SAX contour rings separated by gaps, reproducing the sparse observations of a clinical acquisition. For each case, 2048 query points, locations in space at which the decoder will be asked for a value during training, were sampled near the surfaces and in the surrounding volume, and the signed distance to each surface, or an inside/outside label, was recorded at every one of them; these points provide the dense volumetric supervision for the decoder. The lower sequence of Figure 3.3 connects mesh synthesis in panel (e) and geometric acceptance in panel (f) to the rings in panel (g) and query points in panel (h). Their different spatial distributions distinguish the contours supplied as input from the locations used to supervise predictions throughout the volume. The clinical stream supplies the patient anatomy on which the proposed reconstruction model is evaluated.

3.2.2. Clinical ED and ES surfaces from SAX labels

The clinical data come from two public collections, ACDC [1] and M&Ms-2 [22], which together provide 460 patients (100 from ACDC and 360 from M&Ms-2) with outlines of the cavity, the myocardium, and the RV drawn by experts at ED and ES. They add scanner and site variability, differences in how clinicians delineate the wall, patient-specific and in part pathological anatomy, and the contracted geometry of ES.

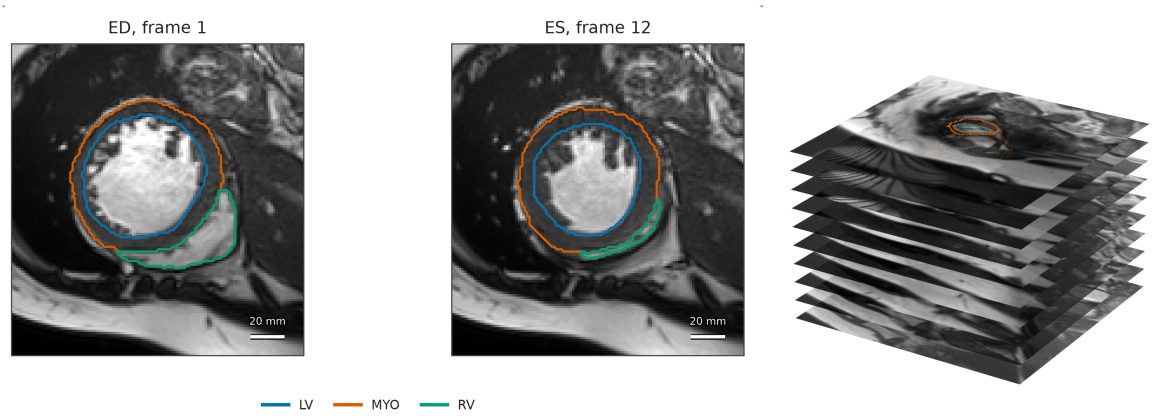

Figure 3.4. Clinical-input view used in the real-data pipeline: (a) representative ACDC ED/ES SAX slices with labels and (b) the corresponding ED SAX stack, with the physical spacing between acquired slices preserved.

In Figure 3.4, the blue and orange outlines identify the endocardial and epicardial boundaries, while green marks the RV. The images on the left show ED and ES; the ED stack on the right preserves the physical spacing between slices. Its gaps are not removed by segmentation, so constructing closed surfaces requires assumptions about the intervening anatomy. The fitting method therefore determines the geometry that the proposed reconstruction model is trained to reproduce.

Three options were available. Connecting corresponding points on consecutive rings (lofting) [48] requires a one-to-one pairing of vertices between rings that the data do not provide, and gives a surface made of flat facets between slices, with a crease at every ring. Fitting the SSM would impose a population prior on every target and, because the available SSM has ED modes only, would either force systolic hearts into a diastolic shape space or treat the two phases differently. Fitting a smooth implicit function to the rings has neither drawback: it needs no vertex pairing and no atlas, and the same procedure applies to ED and ES alike. The clinical targets were therefore built by RBF fitting, which leaves the SSM in its role as the source of synthetic pre-training geometry.

All segmentations are first placed in a common anatomical orientation with their physical voxel spacing preserved, so that every case is expressed in millimetres; the remaining file-handling steps are listed in Table A.2 of Chapter A.

On each acquired SAX slice, the cavity boundary defines the endocardial ring, while the outer boundary of the combined cavity and myocardium defines the epicardial ring. If a label contains several closed boundaries, as can occur with a small detached fragment, only the largest is retained. Each ring is resampled to 60 evenly spaced points and stored at its measured position along the long axis. The blue and orange stacks in Figure 3.5 (c) and (c′) distinguish the endocardial and epicardial constraints. These points serve both as the observations supplied to the encoder and as the constraints used to construct the target surfaces.

The endocardial and epicardial surfaces are fitted separately. For either surface s, let p_i be a contour point and n_i its outward in-plane normal, estimated from the local contour tangent and pointing away from the ring centre. Following Carr et al. [16], a smooth function f_s is defined over the whole volume as a sum of RBFs, smooth bumps whose value depends only on the distance to a centre c_j, plus a linear term, and the surface is taken as its zero level set, the set of points where the function is zero. The thin-plate-spline kernel [49] ϕ used here produces the least bent function that satisfies the constraints, which suits a smooth anatomical wall. The function is

$$f_s(x) = \sum_{j=1}^{K} w_j \phi(\|x - c_j\|_2) + a_0 + a^\top x, \qquad \phi(r) = r^2 \log r, \tag{3.3}$$

In Equation (3.3), the centres c_j are the constraint points, and the weights w_j, a_0, and a are chosen so that the function satisfies the constraints. The contour points fix where the function is zero, while paired points placed a distance $d = 2.5$ mm inside and outside each contour point fix its sign and local scale through

$$f_s(p_i) = 0, \qquad f_s(p_i + dn_i) = d, \qquad f_s(p_i - dn_i) = -d. \tag{3.4}$$

The constraints in Equation (3.4) are applied with mild regularisation, so that the function passes close to, rather than exactly through, the contours and is less sensitive to small irregularities in the manual drawing.

The fitted function is continuous, so a mesh is obtained by sampling it on a fine grid that extends slightly beyond the first and last rings, allowing the surface to close at the base and apex. Light smoothing removes the staircase pattern left by the grid, and marching cubes [24] then traces the zero level set: it visits every cell of the grid and places small triangles wherever the function changes sign, so that the triangles join into a mesh following the surface. Grid spacing and smoothing width are listed in Table A.2. The contour overlays in Figure 3.5 (d) and (d′) locate the acquired observations relative to the fitted surfaces, distinguishing the constrained slice levels from the geometry inferred between them.

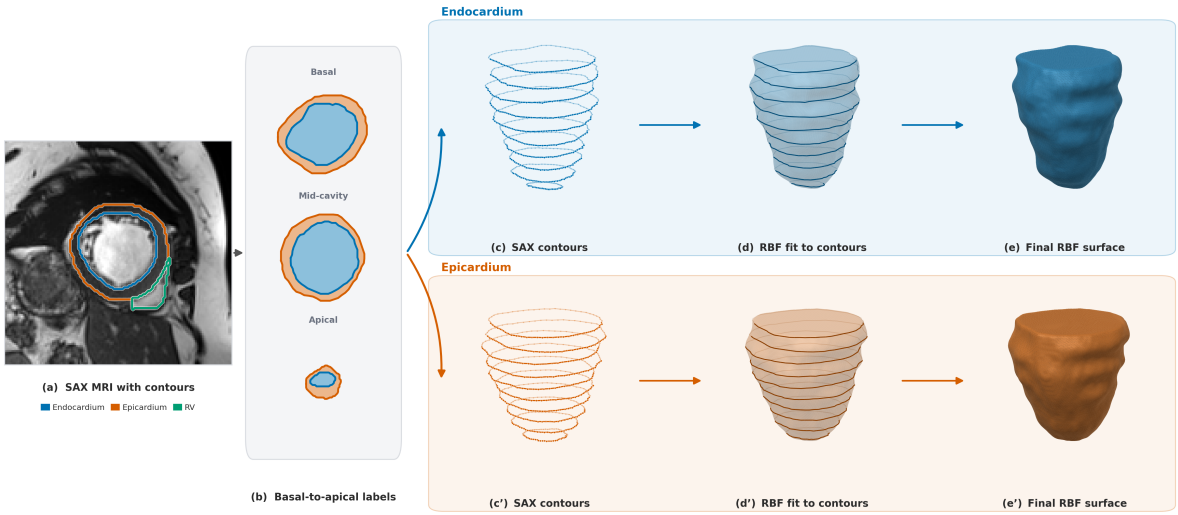

Figure 3.5. Clinical RBF contour-to-surface pipeline. Panel (a) shows one acquired SAX slice with the expert endocardial, epicardial, and right-ventricular contours. Panel (b) shows the corresponding basal, mid-cavity, and apical labels, while panels (c) and (c′) stack the extracted contour rings. Panels (d) and (d′) overlay those constraints on the zero-level surfaces extracted from the smoothed RBF fields, and panels (e) and (e′) show the final topology-checked surfaces used to construct the clinical supervision.

Each extracted mesh must form a single closed surface without holes or handles; minor defects are repaired, and a case is excluded when a surface cannot be recovered, its dimensions are implausible, or the epicardium does not fully enclose the endocardium. The final surfaces in Figure 3.5 (e) and (e′) are the targets that survive these checks.

Both meshes are centred and scaled using the same translation and scale factor. As for the synthetic cases, 2048 query points are sampled near the surfaces and throughout a surrounding box, and their signed distances to both surfaces are recorded.

These fitted surfaces are training targets, not independent anatomical references. Their shape between slices depends on the RBF kernel, off-surface constraints, regularisation, and extraction resolution. Agreement with them therefore measures reproduction of the fitted geometry, not recovery of the true anatomy. Evaluation in Chapter 4 also measures agreement with the observed contours.

3.2.3. Shared sample format, augmentation, and splits

Synthetic and clinical cases come from different sources but must reach the proposed reconstruction model in the same form. A shared sample format lets both branches in Figure 3.1 (a) and (b) feed the same reconstruction stage. Both sources also follow the same augmentation rule, and the data are partitioned by patient.

Table 3.1. Fields of one stored sample, shared by the synthetic and clinical streams, and their role in training.

Field	Origin	Role
Contour points	Synthetic mesh slicing or clinical slice labels	Contours on separated SAX planes supplied to the encoder
Tissue label	Contour extraction	Distinguishes endocardial from epicardial points during tissue-specific pooling
Phase label	Dataset metadata	Conditions the encoder on ED or ES geometry
Surface samples	Prepared endocardial and epicardial meshes	Surface-fitting and wall-offset supervision
Query points and targets	Near-surface and volumetric mesh sampling	Dense signed-distance and off-surface supervision
Mesh metadata	Preprocessing pipeline	Coordinates, scale, faces, and orientation for evaluation and visualisation

Synthetic and clinical cases use the format in Table 3.1, with 10 SAX levels, 60 points per available ring, and 2048 query points per case. Only the contour coordinates and tissue and phase labels are supplied to the encoder. Surface samples and distance targets are used for training, not as reconstruction inputs. The architecture and loss remain unchanged between pre-training and fine-tuning.

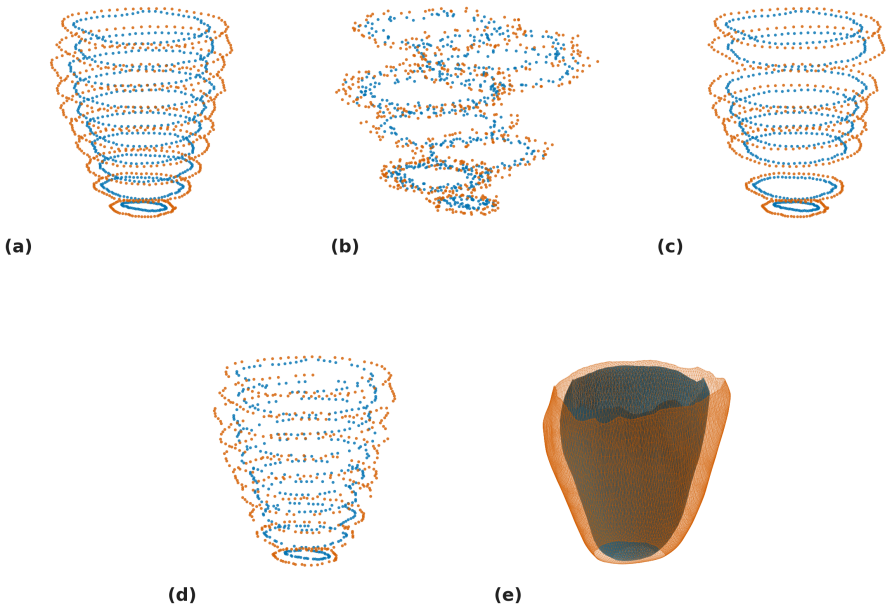

Figure 3.6. Contour-input augmentation: (a) original stack; (b) per-slice translation and point jitter; (c) slice dropout; (d) point dropout; (e) fixed target mesh. Endocardium in blue, epicardium in orange.

The perturbed contour stacks in Figure 3.6 (b) to (d) contrast with the fixed target in panel (e). The model must recover the same geometry despite changes to the observations, which represent acquisition and annotation variability [50, 51]. The modifications include small independent in-plane slice shifts, small random point displacements, global scaling, and point removal. Endocardial and epicardial rings from the same slice are shifted together. Less frequently, whole slices are removed while retaining at least three. Target meshes, surface samples, query points, and tissue and phase labels remain unchanged. Augmentation is applied to both data sources after the patient-level split, preventing modified test cases from entering training.

Not every clinical patient yields a usable target. A study may contain only long-axis views or too few slices, or its segmentation may fail the plausibility checks of Section 3.2.2. After these checks, 444 of the 460 patients provide a valid ED mesh and 423 a valid ES mesh; together with the 1300 accepted synthetic meshes they form the training data summarised in Table 3.2.

Table 3.2. Number of valid samples per data source and cardiac phase.

		Valid samples		
Source	Type	ED	ES	Role
ACDC and M&Ms-2	Clinical MRI	444	423	Fine-tuning and evaluation
UK Digital Heart SSM	Synthetic meshes	1300	—	Pre-training only

The clinical patients are divided into training, validation, and test sets in a 70 %/15 %/15 % proportion, rounded to whole patients and applied to patients rather than to meshes, so that the ED and ES meshes of one patient always fall in the same set and no information passes from the test patients into training [52]. The synthetic meshes are used for pre-training only and never appear in the validation or test sets, so every number reported in Chapter 4 is computed on clinical cases. The role of each stream in the two training stages is summarised in Chapter A. The shared format allows the same reconstruction architecture to process either source.

3.3. Proposed Reconstruction Model

Filling the gaps between slices needs two kinds of information: a sense of the overall ventricular shape, and the detail of the contours closest to each gap. The proposed model gathers both from the contour points in Figure 3.7 (a): a global encoder summarises the whole case, and a local feature grid keeps nearby detail, as shown in panel (b). The implicit decoder in panel (c) uses both to predict the endocardial field and a positive offset that defines the epicardial field, giving the nested surfaces in panel (d). Tissue and cardiac-phase labels allow one network to handle both surfaces and both phases.

Its design builds on established methods for learning from point sets and representing shapes implicitly. Contours are treated as an unordered point set: the same small network

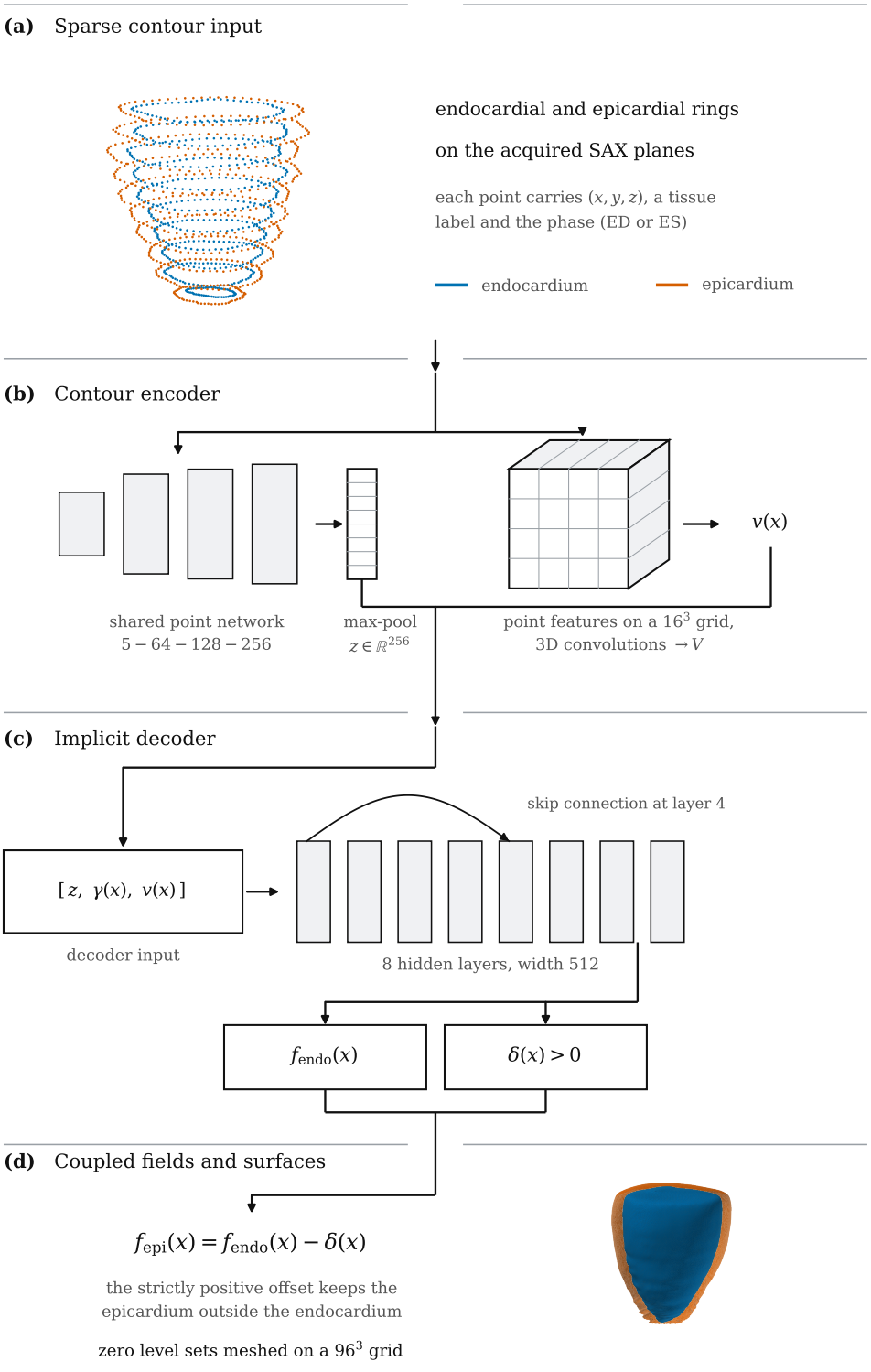

Figure 3.7. Architecture of the proposed reconstruction model, conditioned on cardiac phase, from the contours on separated SAX planes (a) through the global and local encoder paths (b) and the implicit decoder (c) to the coupled fields and extracted surfaces (d). The point cloud in (a) and the cut surfaces in (d) are the observed contours and the corresponding reconstruction of one ACDC case, with the epicardium opened to expose the endocardial cavity. The offset δ shown in the figure is a separation between field values, not a measured thickness.

is applied to every point, and for each learned feature only the strongest response across all points is kept (max-pooling), so the result does not depend on the order in which points are listed, as in PointNet [38]. Spatial detail lost in that summary is recovered from a convolutional feature grid sampled at each query point, following Convolutional Occupancy Networks [53]. The surfaces are zero level sets of signed-distance fields predicted by a network that also receives a compact code describing the whole case (a latent code), in the manner of DeepSDF [17], with Fourier features of the query coordinates [26] and Eikonal regularisation during training [25].

Table 3.3 summarises the adaptations made for LV contours separated by gaps along the long axis and for reconstruction of the paired surfaces. The following subsections define the inputs, encoder, and decoder.

Table 3.3. Components of the proposed reconstruction model, the prior work each one builds on, and what is specific to this work.

Component	Builds on	Specific to this work
Input representation	Unordered point-set input [38]	Every point carries a tissue label and the cardiac phase, allowing joint ED and ES reconstruction
Global encoder	Shared per-point network with max-pooling [38]	Endocardial and epicardial points are pooled separately before being merged into the latent code
Local feature volume	Convolutional feature grid read at the query point [53]	Built from a few hundred contour points averaged into a coarse 16^3 grid; fed to the decoder trunk and to the offset head
Implicit decoder	Latent-conditioned signed-distance network [17] with Fourier features [26]	Latent code supplied by the contour encoder instead of per-shape optimisation
Output coupling	Surfaces as zero level sets of signed-distance fields [17]	Epicardial field defined as the endocardial field minus a strictly positive offset δ, which a wall-distance loss supervises directly
Training	Surface and distance fitting [17]; Eikonal regularisation [25]	Two stages, synthetic SSM pre-training then clinical ED/ES fine-tuning, through one shared sample format

Compared with the recent cardiac implicit models reviewed in Section 2.3.2, which reconstruct from image data [19] or from segmentations of both ventricles on separated slices with long-axis views and a learned biventricular prior [20], the proposed reconstruction model takes

only LV contour points with tissue and phase labels, and it enforces the nesting of the two surfaces by construction.

3.3.1. Inputs to the proposed reconstruction model

The encoder receives the endocardial and epicardial contour stacks in Figure 3.7 (a), with tissue and cardiac-phase labels. MRI intensities and a template mesh are not supplied.

For a batch of B cases processed together, the encoder receives the contour points as the padded tensor defined in Equation (3.5):

$$P \in \mathbb{R}^{B\times N\times 5} \tag{3.5}$$

and a validity mask $M \in \{0,1\}^{B\times N}$ that marks which of the N rows are real points; padding only allows cases with different numbers of points to share a batch. Each valid row is $p_i = (x_i, y_i, z_i, t_i, q_i)$: normalised spatial coordinates, a tissue label t_i that distinguishes endocardial from epicardial points, and a phase label q_i that identifies ED or ES. Attaching the phase to every point, rather than introducing it only at the decoder, lets both encoders interpret the same arrangement of contours differently for a diastolic and a systolic heart. The tissue label serves a different purpose: it allows evidence about the cavity boundary and evidence about the outer boundary to be summarised separately before they are combined.

3.3.2. Contour encoder

A single global summary of the contours captures the overall shape but can lose the local detail needed between slices. The encoder therefore keeps two representations, shown in Figure 3.7 (b): a global summary on the left and a spatial feature grid on the right. The first informs the model about the ventricle as a whole, and the second about the contours nearest to each point.

Following PointNet [38], the global pathway applies a shared network with layer widths $5 \to 64 \to 128 \to 256 \to 256$ and rectified linear unit (ReLU) [54] activations to each point independently. The encoder therefore does not depend on the order of points within a ring or between slices. For the points $P = \{p_i\}_{i=1}^{N}$ of one case, the network assigns each point a 256-dimensional feature, as expressed in Equation (3.6):

$$h_i = \phi_{\text{enc}}(p_i), \tag{3.6}$$

where ϕ_{enc} denotes the shared network.

Unlike standard PointNet, this pooling operation takes separate maxima for the endocardial set $\mathcal{E}$ and the epicardial set $\mathcal{P}$, as shown in Equation (3.7):

$$z_{\text{endo}} = \max_{i\in\mathcal{E}} h_i, \qquad z_{\text{epi}} = \max_{i\in\mathcal{P}} h_i, \tag{3.7}$$

where the maximum is computed independently in each feature channel. This preserves the strongest response to each learned pattern for each boundary before their information is combined. The resulting descriptors form a 512-dimensional vector, which is projected to the global latent code according to Equation (3.8):

$$z = W_z[z_{\text{endo}}, z_{\text{epi}}] + b_z, \tag{3.8}$$

where $z \in \mathbb{R}^{256}$ provides the decoder with a compact description of the complete case.

Compressing all points into z discards where they are. A local pathway, following Convolutional Occupancy Networks [53], keeps this spatial information. A second shared network gives every valid point a 32-dimensional feature, and these features are dropped into a regular 16^3 grid covering the case. Features falling in the same cell are averaged, so that densely sampled regions do not dominate merely because they contain more points.

Three small 3D convolutions then let neighbouring cells exchange information, producing the feature volume in Equation (3.9):

$$V \in \mathbb{R}^{32\times16\times16\times16}. \tag{3.9}$$

For a query point x, the local feature is a weighted average of the eight grid cells surrounding x, with larger weights for the closer cells (trilinear interpolation), as defined in Equation (3.10):

$$v(x) = \text{Interp}(V, x). \tag{3.10}$$

The global code z is shared by every query in a case, whereas $v(x) \in \mathbb{R}^{32}$ describes the contour evidence near x. The grid is deliberately coarse because a few hundred contour points cannot support a fine volume. Its purpose is to provide local context to the decoder, not to define the reconstruction resolution.

3.3.3. Implicit decoder with positive-offset coupling

Each surface is predicted as the zero level set of a field defined over the whole volume, negative inside the surface and positive outside it, in the manner of DeepSDF [17]. With two surfaces to predict, the obvious design gives the decoder in Figure 3.7 (c) one independent output per field. That design leaves the relative position of the two surfaces unconstrained precisely where the reconstruction is least informed.

On an acquired SAX plane, both surfaces are held in place by the observed contours; in the gaps between planes, each is inferred on its own, and nothing prevents them from approaching, touching, or crossing.

Figure 3.8 (a) illustrates the failure. Both surfaces pass through every contour point, yet between two planes the epicardium meets the endocardium on one wall and dips inside it on

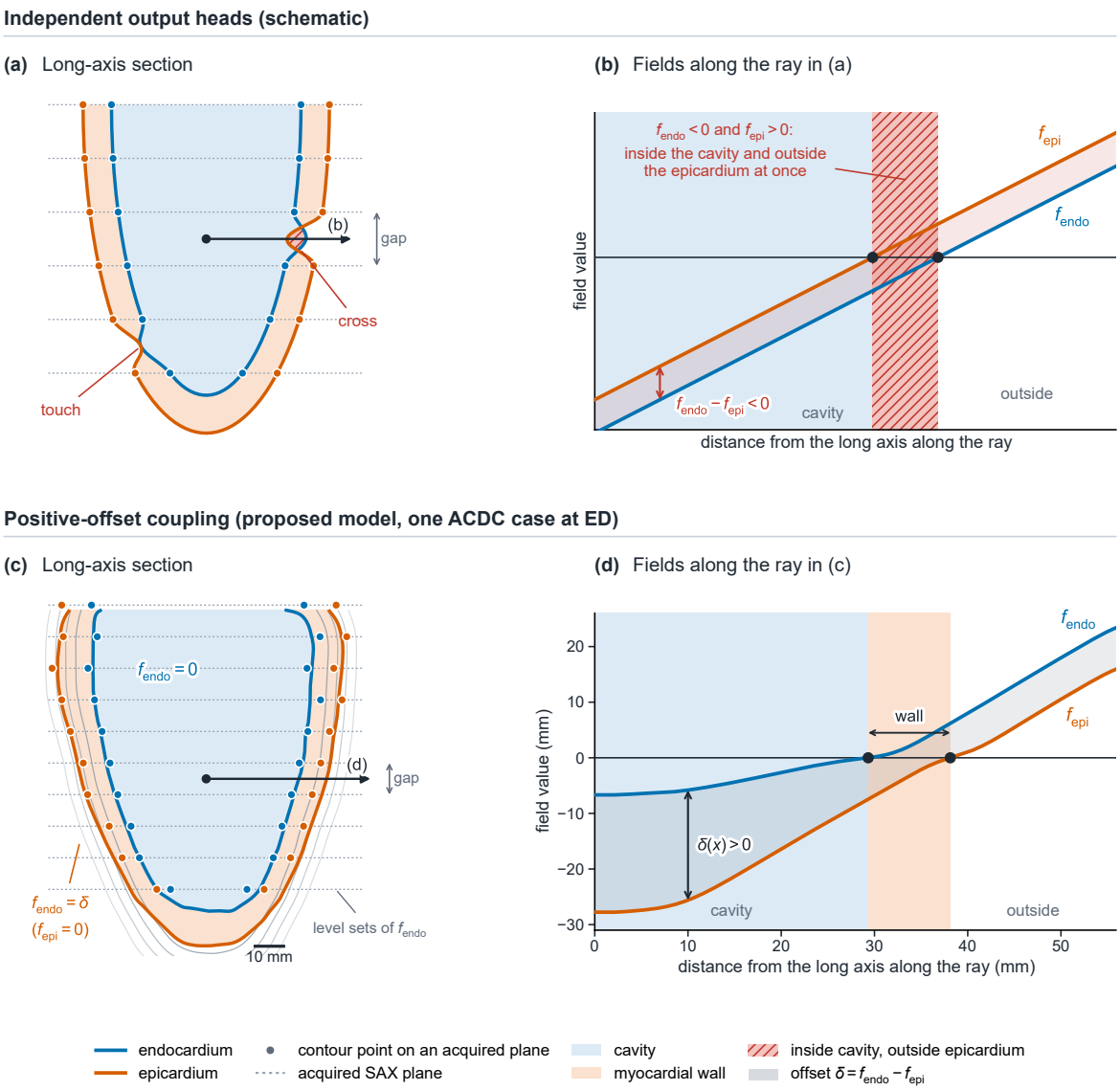

Figure 3.8. Independent and coupled output fields. (a) Schematic of two independently predicted surfaces that match every contour yet touch or cross between SAX planes; (b) fields along the ray in (a). (c) Fields predicted by the proposed reconstruction model for one ACDC case at ED, with the epicardium at $f_{\text{endo}} = \delta$; (d) fields along the ray in (c). Values in (c) and (d) are smoothed with a 2 mm Gaussian for display.

the other. Along a ray through the crossing, Figure 3.8 (b) exposes the resulting contradiction: an interval in which $f_{\text{endo}} < 0$ and $f_{\text{epi}} > 0$, whose points are classified as inside the cavity and outside the epicardium at once, a wall of negative thickness. Because the defect lies away from the observed planes, agreement with the input contours cannot reveal it, and training against complete target surfaces can make it less likely without excluding it for an unseen case. The proposed model therefore excludes it through the structure of its outputs.

Both fields are evaluated by a single coordinate network. A network fed with plain coordinates tends towards overly smooth fields, so each query point is first expanded with Fourier features [26], which add sine and cosine components at several spatial frequencies. For $x \in \mathbb{R}^3$,

the encoding is defined in Equation (3.11):

$$\gamma(x) = [x, \sin(2^0\pi x), \cos(2^0\pi x), \ldots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x)], \tag{3.11}$$

with $L = 3$, giving $3 + 6L = 21$ components. The encoded coordinate is concatenated with the global code to form the decoder input in Equation (3.12):

$$h_{\text{in}}(x) = [z, \gamma(x)] \in \mathbb{R}^{277}. \tag{3.12}$$

The decoder itself is a deep fully connected network of width 512 with smooth activations, which keep the predicted fields smooth and differentiable. Halfway through, the input vector is fed in again (a skip connection, as in DeepSDF [17]), so that the deeper layers still see the case code and the position directly. The local feature $v(x)$ is supplied at several points of the decoder, including the offset head described next, so that the prediction at each position depends on the nearby contours as well as on the global shape. The exact layer configuration is listed in Table A.4 of Chapter A.

The coupling acts at the output stage, where the model departs from an ordinary implicit decoder. The final hidden representation feeds two output layers, or heads: one predicts the endocardial signed-distance value $f_{\text{endo}}(x)$, the other an offset $\delta(x)$ through a final function that can only return positive values. The epicardial field is not predicted separately but derived from both outputs through Equation (3.13):

$$f_{\text{epi}}(x) = f_{\text{endo}}(x) - \delta(x), \qquad \delta(x) > 0. \tag{3.13}$$

The guarantee follows in one step. Wherever $f_{\text{endo}}(x) \leq 0$, that is, in the cavity or on the endocardium, Equation (3.13) gives $f_{\text{epi}}(x) \leq -\delta(x) < 0$, so the point lies strictly inside the epicardium. The interval of Figure 3.8 (b) cannot arise, and the outer surface can neither touch nor cross the inner one, which rules out wall collapse by construction. Geometrically, the epicardium is the set of points where $f_{\text{endo}} = \delta$. It is therefore not a second, independent surface but a positive level set of the endocardial field, whose level varies from point to point with the offset.

Figure 3.8 (c) shows this structure in the fields predicted for an ACDC case whose SAX planes lie 10 mm apart. The epicardium runs among the outer level sets of f_{endo} and remains separated from the endocardium along the whole section, including the gaps in which no contour constrains either surface. On a ray placed halfway between two planes, Figure 3.8 (d) shows the same property in profile: f_{epi} stays below f_{endo} along the entire ray, so the endocardial zero crossing always precedes the epicardial one.

Two qualifications apply. First, the constraint orders the two fields but does not by itself guarantee an anatomically valid surface or a valid mesh once the fields have been sampled on

a grid, which is why the extracted meshes are checked and repaired in Section 3.4.2. Second, δ is not a wall thickness. In Figure 3.8 (d), the offset is the vertical separation between two field values at the same point, whereas the wall is the horizontal separation between the two zero crossings. The offset is also defined throughout the volume: at the centre of the cavity it exceeds 20 mm, although the wall along the same ray spans less than 10 mm. The wall-distance loss of Section 3.4.1 relates δ to the wall only at endocardial points, and thickness is measured on the extracted surfaces with the methods of Section 3.5.

3.4. Training and Inference

Surface targets are needed to learn the reconstruction, but not to use it. In Figure 3.1, the upper branches supply this supervision, whereas the patient sequence below starts from contours alone. Synthetic pre-training and clinical fine-tuning share the same architecture and loss. Once training is complete, the predicted fields are turned into meshes, on which reconstruction quality and wall thickness are evaluated.

3.4.1. Two-stage training and loss

Training proceeds in two stages. Starting with complete synthetic ED surface pairs gives the proposed model a geometric prior before it meets the variability of clinical contours, and the transfer of weights from Figure 3.1 (a) to (b) carries that prior into the clinical stage. The first stage uses only the 1300 synthetic ED cases, whose controlled geometry allows stronger augmentation of the separated contour rings. Fine-tuning then uses the ACDC and M&Ms-2 cases of both phases, adapting the model to clinical contour quality and to systolic anatomy. The patient-level split and the augmentation rule of Section 3.2.3 apply unchanged.

The proposed model is conditioned on cardiac phase rather than trained separately for ED and ES. Because the SSM provides ED geometry only, a separate ES reconstruction model could not have received equivalent pre-training. A shared reconstruction model allows ES cases to benefit from the prior learned on the synthetic ED corpus, while the phase label identifies the cardiac configuration. The RBF-derived targets are constructed identically in both phases, so no second asymmetry is introduced at fine-tuning. The cost is that phase information is coarse, a binary label rather than a position in the cardiac cycle, and that the prior remains biased towards diastolic shape.

The objective combines four established implicit surface losses with a wall loss specific to the coupled outputs of the model. Following Gropp et al. [25], $\mathcal{L}_{\text{surf}}$ guides the reconstructed boundaries towards the target surfaces. Across the sampled volume, $\mathcal{L}_{\text{SDF}}$ matches predicted and target signed distances, adapting DeepSDF supervision [17]. Eikonal regularisation, $\mathcal{L}_{\text{eik}}$, requires each field to change by one unit per unit of distance travelled, the defining property of a true distance, so that the fields behave locally as distances to a surface [25], while $\mathcal{L}_{\text{off}}$

discourages extra surfaces where no target surface exists, following SIREN [55]. To supervise wall separation, $\mathcal{L}_{\text{WT}}$ matches the field offset δ to the target distance between the boundaries, using a scaled Smooth-L1 penalty from Fast R-CNN [56], which is quadratic for small errors and linear for large ones, so that a few large errors do not dominate training.

Formally, let $\mathcal{E}$ and $\mathcal{P}$ denote sampled endocardial and epicardial surface points, $\mathcal{Q}$ the stored query points, $\mathcal{R}$ the union of near-surface and free-space points used for the gradient term, and $\mathcal{F}$ the free-space subset, sampled away from both surfaces. At each $q \in \mathcal{Q}$, $y_{\text{endo}}(q)$ and $y_{\text{epi}}(q)$ are the stored signed-distance targets, and at each $x \in \mathcal{E}$ the wall target $t(x) = -y_{\text{epi}}(x)$ is the distance from that endocardial point to the epicardial surface. Writing $\mathbb{E}$ for the mean over the indicated set of points, the five terms are defined in Equation (3.14):

$$\begin{aligned}
\mathcal{L}_{\text{surf}} &= \mathbb{E}_{x\in\mathcal{E}}[|f_{\text{endo}}(x)|] + \mathbb{E}_{x\in\mathcal{P}}[|f_{\text{epi}}(x)|],\\
\mathcal{L}_{\text{SDF}} &= \mathbb{E}_{q\in\mathcal{Q}}[|f_{\text{endo}}(q) - y_{\text{endo}}(q)|] + \mathbb{E}_{q\in\mathcal{Q}}[|f_{\text{epi}}(q) - y_{\text{epi}}(q)|],\\
\mathcal{L}_{\text{eik}} &= \mathbb{E}_{x\in\mathcal{R}}[(\|\nabla_x f_{\text{endo}}(x)\|_2 - 1)^2] + \mathbb{E}_{x\in\mathcal{R}}[(\|\nabla_x f_{\text{epi}}(x)\|_2 - 1)^2],\\
\mathcal{L}_{\text{off}} &= \mathbb{E}_{x\in\mathcal{F}}[e^{-\alpha|f_{\text{endo}}(x)|}] + \mathbb{E}_{x\in\mathcal{F}}[e^{-\alpha|f_{\text{epi}}(x)|}],\\
\mathcal{L}_{\text{WT}} &= \mathbb{E}_{x\in\mathcal{E}}[\ell_{\beta_{\text{WT}}}(\delta(x) - t(x))],
\end{aligned} \tag{3.14}$$

where $\alpha = 5$ and ℓ_β is the scaled Smooth-L1 penalty with transition parameter β, defined in Equation (3.15):

$$\ell_\beta(a) = \begin{cases} a^2/(2\beta), & |a| < \beta,\\ |a| - \beta/2, & |a| \geq \beta, \end{cases} \qquad \beta_{\text{WT}} = 0.02. \tag{3.15}$$

All distances are expressed in the normalised coordinate system of each case, and the terms are combined into the complete objective in Equation (3.16):

$$\mathcal{L}_{\text{total}} = \lambda_{\text{surf}}\mathcal{L}_{\text{surf}} + \lambda_{\text{SDF}}\mathcal{L}_{\text{SDF}} + \lambda_{\text{eik}}\mathcal{L}_{\text{eik}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{WT}}\mathcal{L}_{\text{WT}}, \tag{3.16}$$

where the weights λ balance the five terms; their values are reported in Table A.3 of Chapter A.

Optimisation uses AdamW [57], a standard gradient-based method, with a learning rate that follows cosine annealing with warm restarts [58]: it decreases along a cosine curve and is reset at intervals, which helps training leave poor solutions. Pre-training lasted approximately 200 epochs (passes over the training set); fine-tuning then ran for 776 epochs, and the checkpoint at epoch 695 was selected by validation loss. Learning rate, batch size, precision, and the remaining settings are collected in Table A.5.

3.4.2. Inference, surface extraction, and comparators

Before reconstruction quality or wall thickness can be assessed, the predicted fields must be turned into meshes, which produces the surfaces in Figure 3.1 (e) from the contours in panel (d).

The encoder computes z and V from the contours once, and the decoder is evaluated on a regular 96^3 grid of points. Marching cubes extracts the zero level sets of f_{endo} and f_{epi} from this grid as the raw endocardial and epicardial meshes [24]. The triangle mesh is therefore an approximation chosen at inference time, not a property of the architecture, and could be extracted at any resolution because the fields themselves are continuous. Small defects in the raw meshes, such as stray fragments or holes, are repaired before evaluation, so the reported metrics describe a cleaned mesh rather than the untouched decoder output.

The proposed model is compared with two reconstructions built from exactly the same contour rings. These comparators embody different and well-understood priors. The first is an RBF implicit fit obtained with the procedure of Section 3.2.2: it interpolates the rings through a smooth field and belongs to the same family as the fine-tuning targets. The second is a fit of the UK Digital Heart Project shape model to the rings. The mean shape is first placed over the rings by matching their long axis, the line through the ring centres, together with their scale and their rotation about that axis. The fit then repeats three steps: each ring point is matched to its nearest SSM vertex, the SSM is re-aligned to the rings by translation, rotation, and scale, and the first 25 mode coefficients are re-estimated while being kept within three SDs. This scheme combines iterative closest point registration [59] and least-squares similarity alignment [60] with the constrained shape update of active shape models [13]. As the published SSM provides ED modes only, this comparator exists at ED alone. Both pass through the same repair and smoothing as the output of the proposed model.

None of the three reconstructions is treated as ground truth. Evaluation compares their distances from the input contours and their pairwise surface and volume agreement. These comparisons assess reconstruction consistency, not the contribution of individual components of the model.

3.5. Wall-Thickness Evaluation

A reconstructed wall has no single local thickness until a rule is chosen for measuring across it. On a curved wall that thins towards the apex, taking the nearest point, travelling perpendicular to the surface, or following a smooth path from one surface to the other can each give a different distance. The colour maps in Figure 3.1 (f) are therefore produced by a separate measurement step applied to the surfaces of panel (e); they are not direct outputs of the model.

Four established methods are compared on the same reconstructed surfaces. The segmentation volume supplies only the contour rings and the orientation of the AHA coordinate system; thickness itself is measured on the reconstructed surfaces. The method whose measurements prove most consistent in Chapter 4 is then used for the regional analysis. The aim is to compare

measurement methods and produce 3D thickness maps, not to assess diagnostic performance for infarction, scarring, or hypertrophy.

3.5.1. Selected thickness methods

Each method estimates the distance across the same wall, but follows a different rule for defining it. The straight distances, curved paths, and ray fan in Figure 3.9 illustrate these choices on a common cross-section. The four methods span the main families identified in Section 2.3.3: distance fields computed on a voxel grid, rays cast from one surface to the other, and methods based on PDEs, which solve an equation over the myocardium to obtain a smooth field running from one surface to the other and then measure along it (Table 3.4). None is assumed in advance to be the best. Each returns a local thickness in millimetres at the endocardial vertices or at points projected onto the endocardium, and the resulting fields are summarised by their mean, median, SD, 5th and 95th percentiles, fraction of valid values, and runtime, and shown as three-dimensional colour maps and as AHA-17 bullseye plots [11], circular diagrams with one cell per segment.

Table 3.4. Wall-thickness methods selected for evaluation on the LV geometry produced by the proposed reconstruction model, with the EDT boundary sum kept as the simplest baseline.

Method	Category	Main measurement principle
Laplace field	PDE	Transmural field lines from a harmonic potential
Yezzi–Prince	PDE	Eulerian transport equations along the transmural field
SDF cone rays	Ray / SDF	Median of multiple cone rays around the surface normal
EDT boundary sum	Volumetric	Sum of endocardial and epicardial distance transforms

The straight arrows from each interior point in Figure 3.9 (a) identify independent distances to the two nearest boundaries. EDT boundary sum adds these distances and is the simplest method. A Euclidean distance transform [43] assigns to every voxel of the myocardium its straight-line distance to the nearest point of a given boundary; the method computes one such transform for each surface and adds them, as defined in Equation (3.17):

$$t^{\mathsf{EDT}}(x) = D_{\mathsf{endo}}(x) + D_{\mathsf{epi}}(x), \tag{3.17}$$

where D_{endo} and D_{epi} are the distance transforms of the endocardium and the epicardium. The sum is exact for two parallel flat walls, but it is limited by voxel size and slightly overestimates thickness where the wall is strongly curved, because the two nearest points do not lie on a common path across the wall.

In Figure 3.9 (b), the curved paths cross dashed lines of equal potential at right angles, defining a direction through the wall rather than two independent nearest-boundary distances.

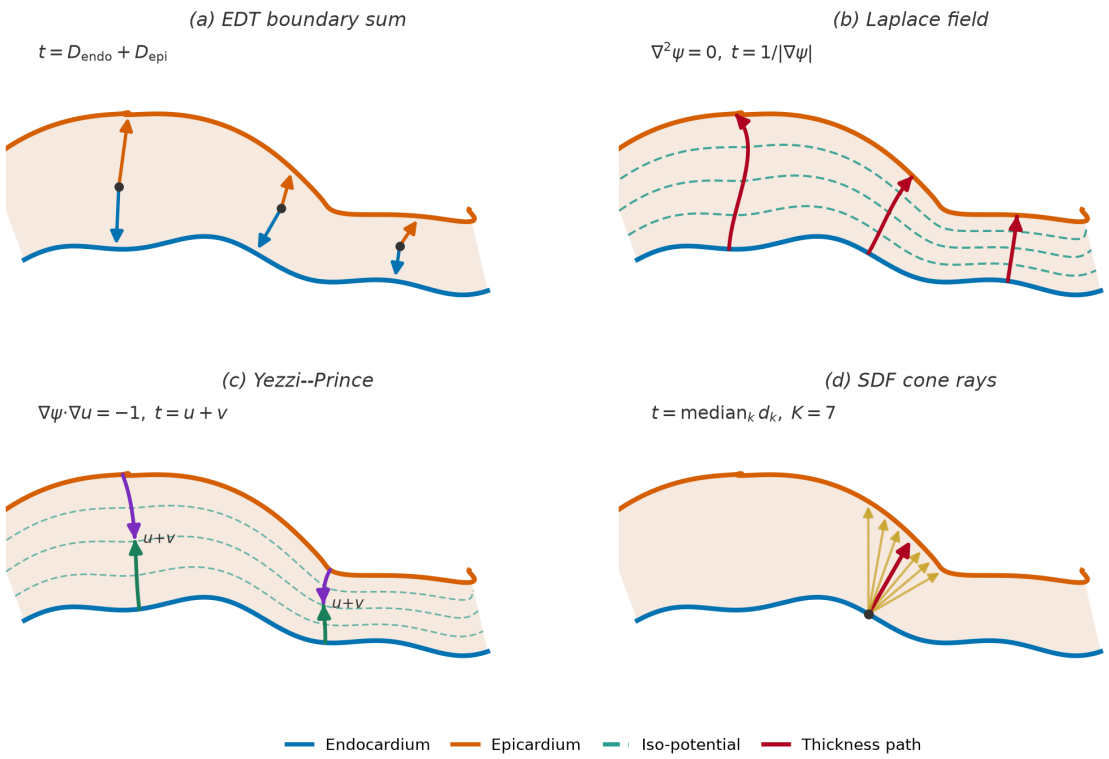

Figure 3.9. Conceptual comparison of the four wall-thickness methods on a single myocardial cross-section (endocardium in blue, epicardium in orange). (a) EDT boundary sum: distances from an interior point to each boundary are added. (b) Laplace field: curved paths of a potential rising from 0 to 1 cross its level lines (dashed) at right angles, and thickness follows from the gradient magnitude. (c) Yezzi–Prince: travel distances u and v from opposite boundaries along the same field are summed. (d) SDF cone rays: the median distance of a fan of rays around the endocardial normal is kept (highlighted), limiting the effect of any single ill-oriented ray.

The Laplace field method constructs this smooth field by defining a scalar potential ψ over the myocardium and solving Laplace's equation with the boundary conditions in Equation (3.18):

$$\nabla^2\psi = 0, \qquad \psi|_{\text{endo}} = 0, \qquad \psi|_{\text{epi}} = 1, \tag{3.18}$$

so that ψ rises smoothly from 0 on the endocardium to 1 on the epicardium, each value being the average of its neighbours (a harmonic field), and acts as a potential whose field lines run across the wall. Thickness is then estimated from the magnitude of the gradient of ψ, the rate at which the field changes across the wall, which is small where the wall is thick and large where it is thin. The paths that follow the gradient never cross one another, which is why the method, introduced for the thickness of the brain cortex by Jones et al. [44], is widely used for layered tissue; its weakness is that the numerical estimate can degrade where the wall is thin or the gradient is small.

The opposing arrows in Figure 3.9 (c) represent distances accumulated from each boundary along the same field. Their sum defines the thickness, rather than estimating it from the gradient magnitude. The Yezzi–Prince method [45] removes this weakness by computing, along the

same field, the distance travelled from each boundary with a pair of transport equations, which accumulate path length step by step while moving along the field and are solved directly on the voxel grid. Their simplified form is given in Equation (3.19):

$$\nabla\psi \cdot \nabla u = -1, \qquad t = u + v, \tag{3.19}$$

where u and v are the distances travelled from the two boundaries and their sum is the thickness.

The ray fan in Figure 3.9 (d) addresses sensitivity to a single surface normal by testing several nearby directions. Each valid ray connects an endocardial point to an epicardial intersection, and the highlighted ray marks the median distance retained. Following the shape diameter function of Shapira et al. [42], which estimates the local thickness of a 3D object from several rays cast into its interior, K rays are cast within a cone of half-angle 30° around the normal, and the median of the valid intersection distances is kept, as defined in Equation (3.20):

$$t_i^{\text{cone}} = \text{median}_{k=1}^{K}\, d_{ik}, \qquad K = 7. \tag{3.20}$$

As a result, the estimate remains approximately transmural, measuring across rather than along the wall, while being more robust than a single normal ray.

3.5.2. Local three-dimensional wall thickness

Thickness is retained at every endocardial vertex of the repaired meshes of Section 3.4.2 so that local variation remains visible before regional averaging.

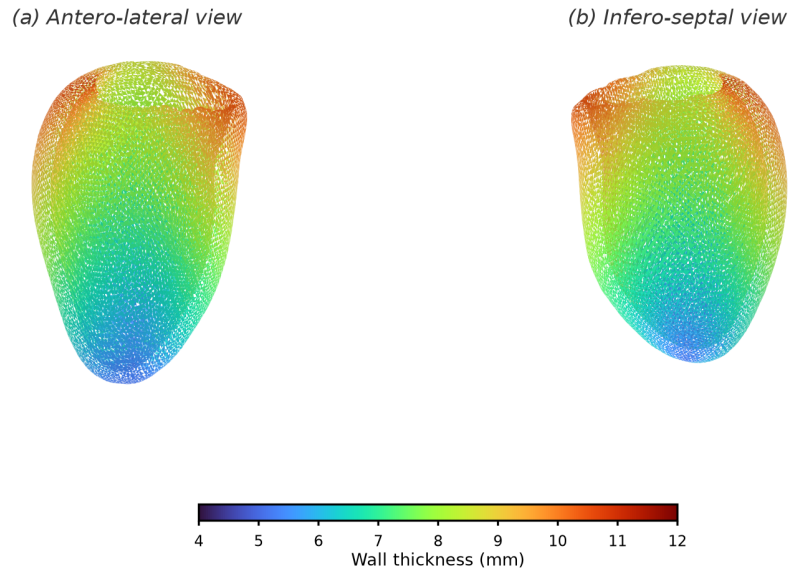

Figure 3.10. Local three-dimensional wall-thickness map on the reconstructed LV endocardial surface, shown from two opposing viewpoints. Each vertex is coloured by its local wall thickness..

The opposing views in Figure 3.10 expose different parts of the same surface map, rather than limiting inspection to selected slices. Colour encodes thickness in millimetres: in the illustrated example, the transition from cooler apical colours to warmer basal colours indicates the change from a thinner to a thicker wall. This local map is the primary output of the measurement stage, preserving features that an AHA segment average could conceal. A fixed colour scale supports comparisons across patients and phases.

3.5.3. Regional AHA-17 mapping

Comparing local thickness across patients requires dividing every ventricle into the same anatomical regions. The AHA 17-segment model [11] provides this partition for clinical reporting, linking the surface patches in Figure 3.11 to matching bullseye cells. Their colours identify segments rather than thickness values, allowing the local map from the selected method to be summarised within each region. This segmentation scheme was defined for two-dimensional imaging and divides the ventricle along its long axis into basal, mid-cavity, and apical thirds. A representative SAX slice from each third is partitioned into angular sectors around the cavity: six sectors in the basal and mid-cavity regions and four in the apical region. The sectors are numbered from the anterior wall, with the septal segments facing the RV, while the apical tip forms the seventeenth segment. In routine use, an observer measures the thickness of each segment on the corresponding slice.

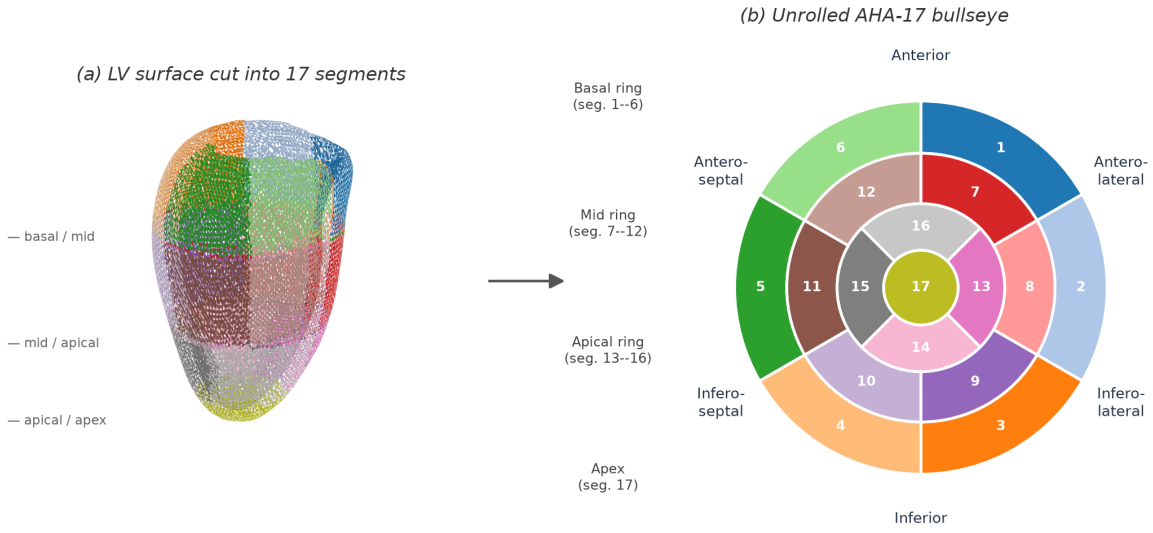

Figure 3.11. AHA-17 partitioning of the reconstructed LV surface and corresponding bullseye representation.

Transferring this definition to a reconstructed surface requires only that every vertex receive the two coordinates the definition relies on, a position along the long axis and an angle around it; the segments then become patches of the three-dimensional surface rather than readings on

single slices. The long axis is taken as the direction of the SAX stack. The base and apex are the first and last slices that contain myocardium, the base being the end whose cavity cross-section is larger, and every endocardial vertex is given a normalised position between 0 at the base and 1 at the apex. Vertices in the first third of this range form the basal ring, those in the second third the mid-cavity ring, those between two thirds and 0.85 the apical ring, and those beyond 0.85 the apical cap, which is segment 17.

The angle of each vertex is measured in the SAX plane around the centre of the endocardial vertices. Its orientation is fixed by the right-ventricular label: the direction from the LV centre to the right-ventricular centre identifies the septum, and the anterior reference lies a quarter turn from it. The basal and mid-cavity rings are divided into six sectors of $60°$, while the apical ring is divided into four sectors of $90°$. These divisions form segments 1–6, 7–12, and 13–16, respectively (Figure 3.11).

Before assigning the segments, a quality-assurance view displays the detected base, apex, and septal direction. The rings are then unrolled onto concentric circles to form the bullseye, with the basal ring outermost and the apical cap at the centre. Each cell therefore summarises the vertices of one surface patch. Within each segment, the mean and percentiles provide complementary regional summaries rather than relying on a single statistic. These summaries complement rather than replace the per-vertex maps.

CHAPTER 4

Results and Evaluation

This chapter presents the experimental evaluation of the approach described in Chapter 3.

4.1. Experimental Setup

The evaluation uses the final version of the proposed model, which was pre-trained on the synthetic SSM meshes and then fine-tuned on the real ED and ES data, as described in Chapter 3. Datasets and patient-level splits are reported in Section 3.2.3. General reconstruction and wall-thickness evaluations use the held-out test patients from both ACDC and M&Ms-2. The analysis by disease group uses a separate ACDC subset of 30 test patients, comprising 20 normal (NOR) and 10 HCM cases. The HCM group is included so that the proposed model can be examined on abnormal as well as normal LV geometry. No independent 3D ground truth is available, so the following results describe contour fidelity and agreement between methods rather than reconstruction accuracy. Three questions guide the evaluation: how faithfully the proposed model follows the observed LV contours and how far it agrees with alternative reconstructions of them, whether it represents the ED–ES change while maintaining a valid wall, and how closely the resulting thickness measurements agree with those obtained from an independent method on the same input.

4.1.1. Metrics

Reconstruction geometry is described with surface and volume measures taken from the review of Taha and Hanbury [21] and from common practice in cardiac segmentation [1] and implicit-surface learning [18]. They are used between a reconstruction and the contour rings it was built from, where they show how closely the surface follows the observations, and between two reconstructions of the same rings, where they show how much the methods differ without blaming either one.

The *contour distance* is the gap from each observed ring point to the nearest point of the reconstructed surface, summarised by its mean and upper percentile; it only checks the acquired slices, not the surface between them. The *Chamfer distance* [17] averages the nearest-point gaps between two surfaces in both directions, and *HD95*, the Hausdorff distance taken at its upper percentile [21], reports the large gaps that this average hides. Both are given in millimetres for the endocardium and the epicardium.

The *Dice similarity coefficient* (DSC) [21] compares volumes rather than surfaces: the cavity inside the endocardium and the myocardium between the two surfaces, each filled on a shared voxel grid. It is the shared volume relative to the total volume of both reconstructions, rising from no overlap to complete overlap; these volumes come from the reconstructed meshes, not from independent 3D annotations. Enclosed *volumes* are also reported in millilitres for each method separately, the epicardial value including both cavity and wall, and the *watertight rate*, the fraction of meshes closed without holes, is a requirement for these volumes to exist rather than a score. The average symmetric surface distance, intersection over union, F-score and normal consistency are left out because they repeat the Chamfer distance or the DSC, depend on a chosen tolerance, or measure orientation rather than position.

Wall thickness is summarised by the mean, SD, and lower and upper percentiles across the myocardial surface, in millimetres; the SD describes how much the thickness varies over one wall and the percentiles mark its thin and thick ends without relying on single extreme values. Between phases, the absolute thickening $\Delta_{\mathrm{ES}-\mathrm{ED}}$ is the ES minus the ED thickness and the relative thickening is that increase as a percentage of the ED value.

Agreement between wall-thickness measurements follows Bland and Altman [61] and uses paired values, either patient means or measurements at the same surface points. The *Pearson correlation coefficient* r shows whether two methods rank the same patients or locations as thicker, not whether their values match. The *mean absolute error* (MAE) is the average size of the difference in millimetres, and the *root-mean-square error* (RMSE) gives more weight to large differences. The Bland–Altman *bias* is the mean signed difference, showing whether one method reads consistently higher, and the *limits of agreement* (LoA) give the range expected to contain most individual differences. The two-way *intraclass correlation coefficient* (ICC) rates agreement against the spread of the paired values and, unlike r, penalises a constant offset; it is read against the thresholds of Koo and Li [62].

4.1.2. Methods Used for Comparison

The reconstruction is compared with two alternative reconstructions of the same input contour rings: a radial-basis-function implicit fit, which interpolates the rings through a smooth field, and a fit of the public SSM, which keeps the reconstructed shape within the variation represented by the SSM. Differences between their results show how assumptions about shape between slices affect the reconstructed geometry. Both use the same rings as the proposed model and undergo the same mesh repair and measurement steps, as described in Section 3.4.2, so their numbers are directly comparable. They are evaluated only at ED: the public SSM is distributed with ED modes only.

The RBF fit is used for comparison, not as ground truth. It belongs to the same family as the surfaces fitted to the contours for training, so close agreement with the proposed model is partly expected and is not independent confirmation. By contrast, the SSM fit is independent of the training targets, but some patients have shapes outside the variation represented by its training population. Wall thickness is compared between the proposed model, the RBF fit, and the SSM fit with the four algorithms selected in Section 3.5.1: the Laplace field method, the Yezzi–Prince method, the SDF cone-ray method, and the EDT boundary sum. Because the SAX stacks are sparse relative to their in-plane resolution, all three reconstruction methods must infer the through-plane geometry, and their absolute millimetre values should be read with care.

4.2. Training

Training proceeds in two stages. The first pre-trains the signed-distance decoder for roughly 200 epochs on the synthetic SSM meshes. This stage provides a geometric prior, and its checkpoint initialises the second stage. The proposed model is then fine-tuned on the real ED and ES data described in Chapter 3. The available training history records the fine-tuning stage; no numerical claim is made about the pre-training loss because its per-epoch log was not retained.

Fine-tuning runs for 776 epochs with the AdamW optimiser (learning rate 2×10^{-5}, weight decay 5×10^{-4}) under a cosine-annealing schedule with warm restarts, mixed-precision arithmetic, and an effective batch size of 32.

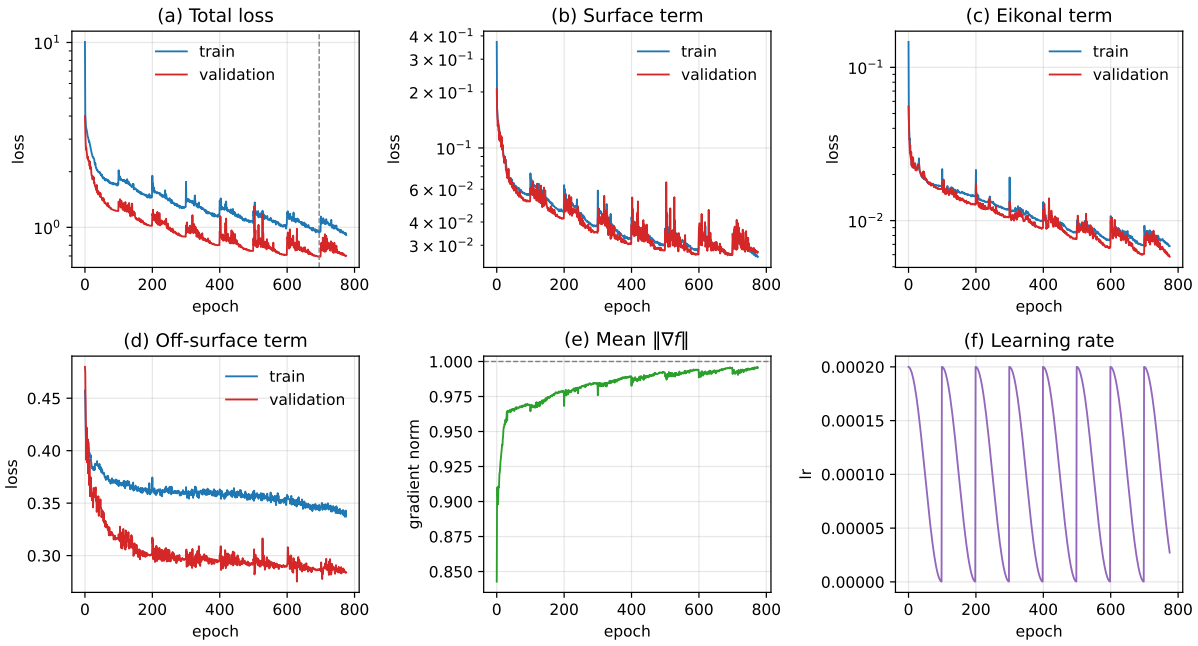

Figure 4.1. Training history. (a)–(d) Total loss and its surface, Eikonal, and off-surface components for the training and validation splits; the dashed line marks the best epoch. (e)–(f) Mean gradient norm (target one) and cosine-annealing learning rate. Synthetic pre-training is not shown.

The composite objective is the five-term loss of Equation (3.16), combining surface, dense SDF, Eikonal, off-surface, and wall-thickness supervision. Figure 4.1 plots the total loss and its principal components together with two optimisation diagnostics. The total training loss decreases from 10.07 at the first fine-tuning epoch to 0.91, and the validation loss settles at 0.70, with the best checkpoint selected at epoch 695 before mild over-fitting appears. The surface and Eikonal terms fall by more than an order of magnitude, indicating successful optimisation of the surface-fitting and Eikonal objectives. Consistently, the mean gradient norm $||\nabla f||$ stabilises at 0.996, close to the unit value encouraged by the Eikonal term.

4.3. Results

Three analyses cover reconstruction consistency, wall-thickness measurement, and regional analysis with the AHA 17-segment model.

4.3.1. Reconstruction Consistency

All values below are from the held-out ACDC and M&Ms-2 test cohort.

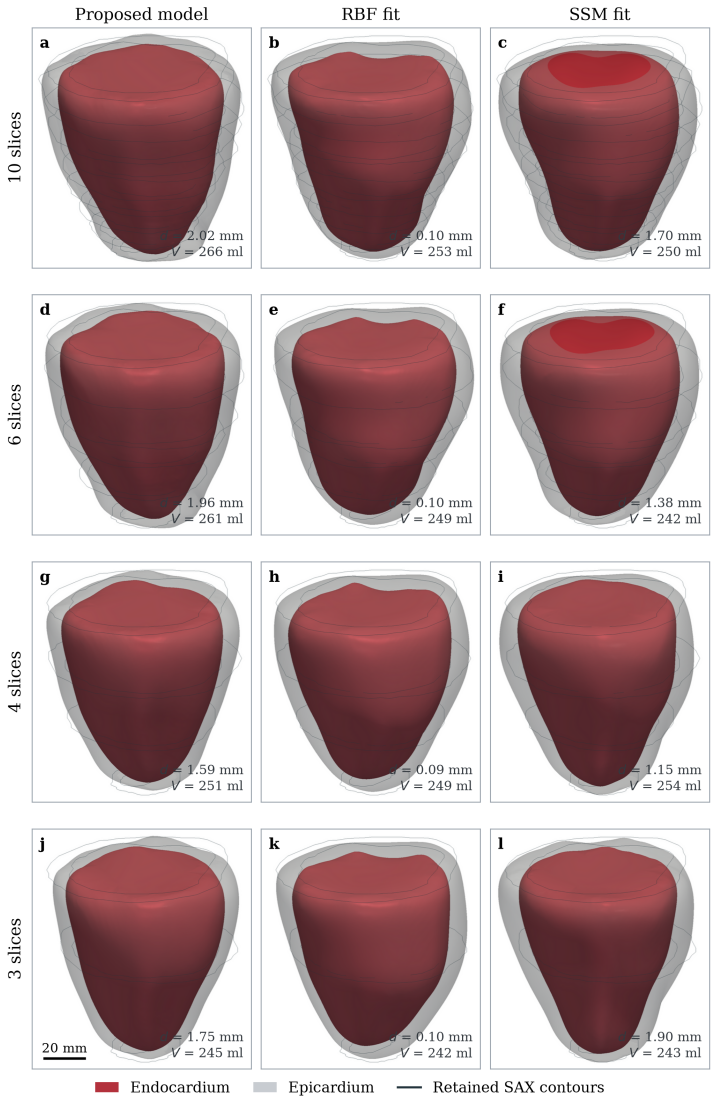

Figure 4.2. Left-ventricular surfaces of one ACDC case reconstructed from an ED SAX stack with ten, six, four, and three retained slices (rows) by the proposed model, RBF fit, and SSM fit (columns). Red: endocardium; grey: epicardium; dark rings: input contours.

Figure 4.2 shows one case reconstructed by the three methods. In the complete ten-slice row, the three surfaces share the same overall ventricular shape. The reduced-slice rows are analysed in Section 4.3.2.

Table 4.1 reports the distance from each input contour ring to the reconstructed surface and the enclosed volumes, the endocardial value being the cavity alone and the epicardial value the cavity together with the wall.

Table 4.1. Distance from the input SAX contour rings to each reconstructed surface, and enclosed volumes, at ED over the held-out test cohort (mean ± SD of per-patient values; p_{95}: mean per-patient 95th percentile). Bold marks the lowest distance.

Method	Surface	Contour distance (mm)	p_{95} (mm)	Volume (ml)
Proposed model	Endocardium	1.50 ± 0.23	3.56	119.3 ± 31.3
	Epicardium	1.96 ± 0.28	5.23	255.3 ± 76.7
RBF fit	Endocardium	**0.13 ± 0.01**	**0.30**	118.6 ± 30.2
	Epicardium	**0.11 ± 0.01**	**0.26**	233.5 ± 71.4
SSM fit	Endocardium	1.82 ± 0.67	5.51	126.9 ± 32.2
	Epicardium	1.91 ± 0.76	6.23	215.2 ± 56.3

The RBF fit follows the rings almost exactly, at 0.13 mm on the endocardium and 0.11 mm on the epicardium, because it passes through the contour points by construction. The proposed model lies 1.50 and 1.96 mm from the same rings, which is close to the in-plane pixel size of 1.37 to 1.88 mm. The SSM fit has similar mean distances, 1.82 and 1.91 mm, but a wider spread between patients: its per-patient endocardial distance ranges from 0.92 to 3.44 mm, against 1.20 to 2.14 mm for the proposed model.

In the 30-patient ACDC pathology subset, the mean contour distance of the SSM fit is 1.42 mm for normal cases and 2.63 mm for HCM cases. The corresponding distances for the proposed model are 1.45 and 1.60 mm.

Cavity volumes are similar for the proposed model, the RBF fit and the SSM fit, at 119, 119 and 127 ml, respectively. Myocardial volumes, obtained as the epicardial minus the cavity volume, differ more, at 136, 115 and 88 ml.

Table 4.2 compares the three reconstructions with one another, by the distance between their boundaries and by the overlap of the cavity and wall that those boundaries enclose.

Table 4.2. Pairwise agreement between the three reconstructions of the same ED contour rings over the held-out test cohort (mean ± SD of per-patient values). Entries are symmetric disagreements, not errors against a reference. Bold marks the closest pair.

Pair	Metric	Endocardium/cavity	Epicardium/myocardium
Proposed model vs. RBF fit	Chamfer (mm)	**1.57 ± 0.18**	**1.90 ± 0.24**
	HD95 (mm)	**4.24 ± 0.86**	**6.07 ± 0.94**
	DSC	**0.91 ± 0.02**	**0.77 ± 0.05**
Proposed model vs. SSM fit	Chamfer (mm)	2.08 ± 0.51	2.59 ± 0.73
	HD95 (mm)	5.88 ± 1.91	6.72 ± 2.32
	DSC	0.88 ± 0.03	0.69 ± 0.04
RBF fit vs. SSM fit	Chamfer (mm)	2.01 ± 0.67	2.05 ± 0.64
	HD95 (mm)	5.88 ± 2.95	6.62 ± 2.67
	DSC	0.89 ± 0.04	0.76 ± 0.06

The proposed model and the RBF fit form the closest pair on every metric and surface, with Chamfer distances of 1.57 mm on the endocardium and 1.90 mm on the epicardium and a myocardial DSC of 0.77. Both pairs involving the SSM fit are further apart, at 2.08 and 2.01 mm on the endocardium.

Across the three pairs, the endocardial Chamfer distance ranges from 1.6 to 2.1 mm. Cavity DSC is uniformly high, at 0.88 to 0.91, whereas myocardial DSC is lower, at 0.69 to 0.77, and falls to 0.58 for the worst patient in a comparison involving the SSM fit.

4.3.2. Slice Sensitivity

Each patient's complete ED stack is reduced to six, four and three slices. The most basal and most apical slices are always kept, so the base-to-apex extent stays the same and only the number of slices within it changes. Because the acquired stacks differ in length, a patient contributes to a given level only when its complete stack has more slices than that level. The lower rows of Figure 4.2 show the effect on one case.

Table 4.3. Deviation of each method's reduced-slice reconstruction from its complete-stack reconstruction at ED on the held-out test cohort. Bold marks the smallest deviation at each slice level.

		Chamfer (mm)		DSC		
Method	**Slices**	**Endo.**	**Epi.**	**Cavity**	**Myoc.**	ΔV_{myo} **(%)**
Proposed model	6	**0.79**	0.87	**0.971**	**0.891**	−0.4
	4	1.27	1.48	0.953	0.819	−2.1
	3	1.83	2.17	0.934	0.782	−3.8
RBF fit	6	0.83	**0.83**	0.968	0.887	**0.0**
	4	**1.15**	**0.88**	**0.959**	**0.858**	**−1.8**
	3	**1.45**	**1.32**	**0.947**	**0.805**	−6.2
SSM fit	6	0.89	0.95	0.968	0.883	−1.8
	4	1.46	1.39	0.942	0.812	−3.8
	3	1.94	1.92	0.921	0.775	**+2.6**

Table 4.3 compares each reduced-stack reconstruction with the one the same method produces from the complete stack, so the values show the effect of removing slices rather than the differences between methods; ΔV_{myo} is the change in wall volume relative to that complete-stack result. All three methods still produce closed surfaces from three slices.

With six slices, the reconstructions change very little: surface distances stay below 1 mm and myocardial DSC is close to 0.89 for every method. With fewer slices, the RBF fit changes least, while the proposed model and the SSM fit change by similar amounts. The difference is clearest on the epicardium, where the Chamfer distance at three slices reaches 2.17 mm for

the proposed model, 1.92 mm for the SSM fit and 1.32 mm for the RBF fit. Cavity DSC stays above 0.92 throughout, and myocardial DSC at three slices lies between 0.775 and 0.805.

Myocardial volume changes by only a few percent. At three slices, the proposed model and the RBF fit lose 3.8% and 6.2%, whereas the SSM fit gains 2.6% after losing volume with six and four slices, so its small change at that level does not mean that it follows the complete-stack volume more closely.

4.3.3. Wall-Thickness Measurement

The four selected methods are applied to the proposed reconstruction, the RBF fit, and the SSM fit for every patient at ED. Table 4.4 reports, for each method and reconstruction, the cohort average of the per-patient mean, SD, and 5th and 95th percentiles, the SD describing variation across each patient's wall rather than between patients.

Table 4.4. Wall-thickness statistics at ED over the held-out ACDC and M&Ms-2 test cohort. Each entry is the cohort average of the corresponding per-patient statistic, in millimetres.

Method	Geometry	Mean (mm)	SD (mm)	p_5 (mm)	p_{95} (mm)
Laplace field	Proposed model	9.18	2.14	4.75	12.20
	RBF fit	8.36	2.21	5.03	12.29
	SSM fit	6.48	1.38	3.87	8.43
Yezzi–Prince	Proposed model	10.01	1.64	7.41	12.74
	RBF fit	8.86	2.15	5.57	12.71
	SSM fit	7.10	1.23	4.75	8.88
SDF cone rays	Proposed model	12.47	4.87	8.01	20.68
	RBF fit	9.38	2.49	5.63	13.80
	SSM fit	7.11	1.50	4.12	9.13
EDT boundary sum	Proposed model	9.55	1.66	6.62	12.17
	RBF fit	8.51	2.00	5.42	12.05
	SSM fit	7.02	1.22	4.65	8.77

The three non-ray methods agree closely on each reconstruction, with mean thicknesses of 9.18 to 10.01 mm on the proposed reconstruction, 8.36 to 8.86 mm on the RBF fit and 6.48 to 7.10 mm on the SSM fit. The SDF cone-ray method stands apart on the first two, at 12.47 and 9.38 mm, and also spreads more widely: on the proposed reconstruction, its per-patient SD is 4.87 mm, against 1.64 to 2.14 mm for the other methods.

Table 4.5 compares patient-level mean thickness on the proposed reconstruction with the two fitting baselines, with the measurement algorithm held fixed so that differences reflect the geometry alone.

Table 4.5. Patient-level agreement between wall thickness on the proposed reconstruction and on each fitting baseline at ED. Bias is the proposed-model value minus the baseline value. Bold marks the closest agreement in each column.

	Proposed model vs. RBF fit			Proposed model vs. SSM fit		
Method	**Bias (mm)**	**RMSE (mm)**	r	**Bias (mm)**	**RMSE (mm)**	r
Laplace field	**+0.82**	**0.86**	**0.99**	+2.70	3.13	0.72
Yezzi–Prince	+1.16	1.23	**0.99**	+2.91	3.49	**0.77**
SDF cone rays	+3.10	3.81	0.95	+5.36	6.64	0.67
EDT boundary sum	+1.03	1.10	**0.99**	**+2.53**	**3.04**	**0.77**

For every measurement method, patient-level mean thickness is lower on the RBF and SSM fits than on the proposed reconstruction, with larger offsets for the SSM. Patient rankings agree more closely with the RBF fit.

Table 4.6 compares the three other methods with the Laplace field at the same surface points on the proposed reconstruction, so that only the measurement algorithm changes.

Table 4.6. Agreement of each wall-thickness method with the Laplace field over the myocardial surface points of the held-out test cohort. Bias is the method value minus the Laplace-field value. The Laplace field is the reference for comparison, not ground truth.

Method	r	**MAE (mm)**	**RMSE (mm)**	**Bias (95% LoA) (mm)**	**ICC**
Yezzi–Prince	**0.80**	0.89	2.14	+0.83 (±3.86)	**0.77**
SDF cone rays	0.44	3.38	7.29	+3.31 (±12.74)	0.28
EDT boundary sum	0.78	**0.81**	**2.07**	**+0.37** (±3.99)	**0.77**

The EDT boundary sum differs from the Laplace field by 0.81 mm on average, with a bias of +0.37 mm, and Yezzi–Prince by 0.89 mm, with a bias of +0.83 mm; both reach ICC $= 0.77$, which corresponds to moderate to good agreement under the thresholds of Koo and Li [62]. The cone-ray method differs by 3.38 mm on average, with a bias of +3.31 mm, LoA of ±12.74 mm and ICC $= 0.28$, which corresponds to poor agreement.

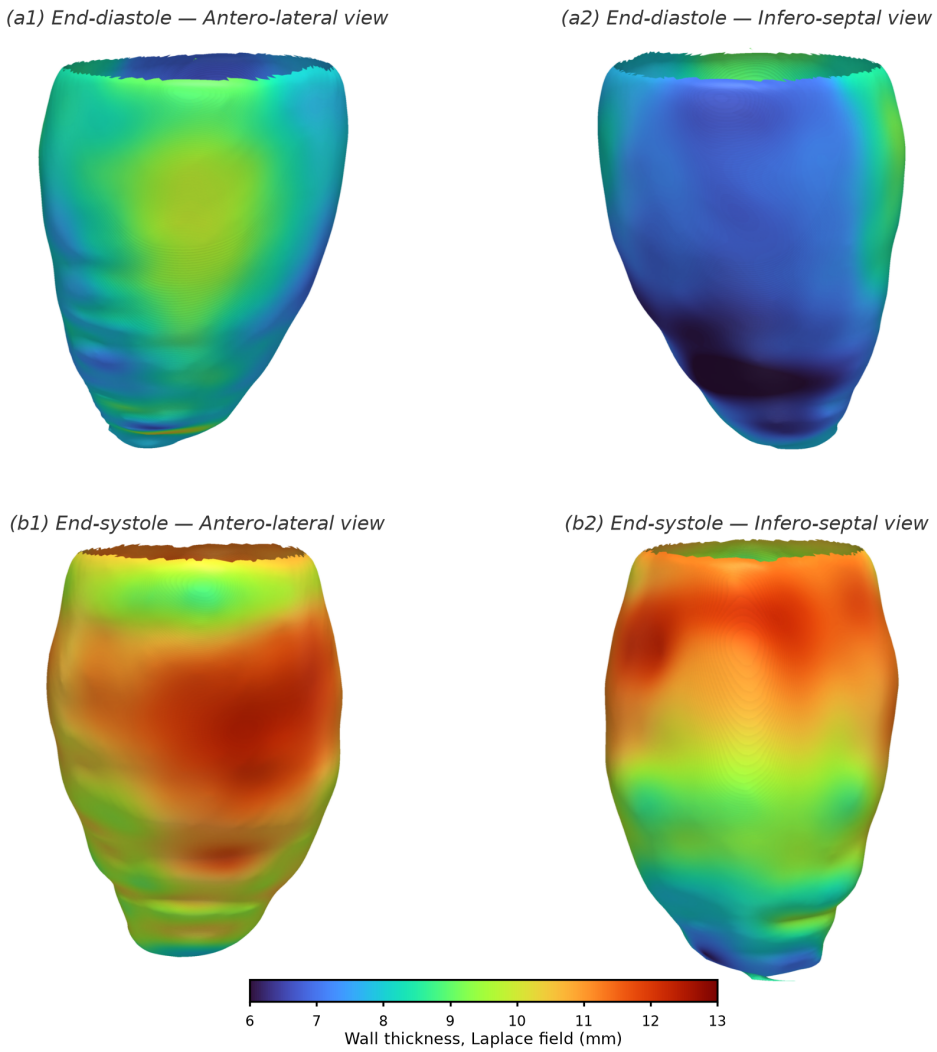

Figure 4.3. Predicted wall thickness on the reconstructed surface of one ACDC case at ED (top) and ES (bottom), shown from two viewpoints on a shared colour scale. Values are scaled to the input segmentation.

Figure 4.3 shows the predicted wall thickness as a heatmap on the reconstructed surface. The values are scaled to the input segmentation, so the map shows where the wall is thicker or thinner rather than its absolute thickness. The circumferential bands follow the spacing of the SAX slices.

4.3.4. Regional Analysis over the AHA 17-Segment Model

Regional results are mean Laplace-field thicknesses [44] measured directly on the meshes within the AHA segments [11], using the mapping described in Section 3.5.3. They are not the segmentation-scaled values of Figure 4.3.

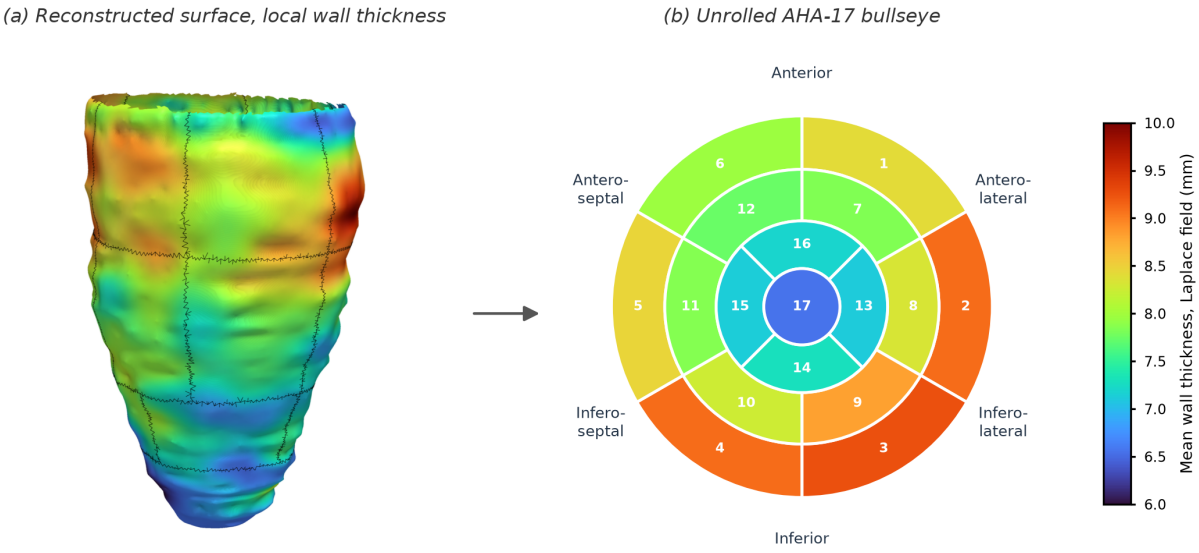

Figure 4.4. AHA 17-segment analysis of the representative case at ED: (a) reconstructed surface coloured by the mean Laplace-field wall thickness of each segment; (b) the same values on the standard bullseye, with basal (1–6), mid-cavity (7–12) and apical (13–16) rings and the apical cap (17).

Figure 4.4 shows these regional means on the reconstructed surface and the corresponding bullseye for the representative case.

Table 4.7 reports the regional pattern over the 30-patient ACDC pathology subset and its change between phases. Entries are segment means averaged across patients; variation between patients is examined in Table 4.9.

Over the ACDC pathology subset the reconstructed wall averages 9.21 mm at ED and 13.87 mm at ES, corresponding to 50.6% relative thickening. At ED the septal mid-cavity segments are the thickest, while the apical cap is the thinnest at 7.76 mm. Every segment thickens between phases; the smallest relative change is 32.1% in the basal anteroseptal segment and the largest is 64.6% in the mid inferolateral segment. No fitting baseline is available at ES, so these phase values describe the model output only.

Table 4.7. Regional AHA 17-segment wall thickness on the reconstructed geometry over the 30-patient ACDC pathology subset.

#	Segment	ED (mm)	ES (mm)	△ (%)
1	Basal anterior	9.99	13.71	+37.1
2	Basal anteroseptal	10.15	13.40	+32.1
3	Basal inferoseptal	9.89	13.43	+35.7
4	Basal inferior	9.09	13.81	+51.9
5	Basal inferolateral	8.93	14.27	+59.8
6	Basal anterolateral	9.53	13.87	+45.5
7	Mid anterior	9.19	13.79	+50.0

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Table 4.7. AHA 17-segment thickness over the ACDC pathology subset *(continued).*

#	Segment	ED (mm)	ES (mm)	Δ (%)
8	Mid anteroseptal	10.37	14.68	+41.6
9	Mid inferoseptal	10.19	15.05	+47.6
10	Mid inferior	9.25	14.41	+55.8
11	Mid inferolateral	9.09	14.96	+64.6
12	Mid anterolateral	8.99	14.17	+57.6
13	Apical anterior	8.50	13.63	+60.3
14	Apical septal	8.77	13.86	+58.1
15	Apical inferior	8.44	13.29	+57.5
16	Apical lateral	8.43	13.78	+63.6
17	Apex	7.76	11.69	+50.7
Overall		**9.21**	**13.87**	**+50.6**

Note. Values are means for the 30 ACDC test patients used in the pathology analysis, measured with the Laplace field. Δ is the relative thickening from ED to ES.

Table 4.8 separates the normal and HCM patients, and both groups show an internally coherent regional pattern. In the normal cases the ED thickness varies little around the base and the mid-cavity, from 7.92 to 8.81 mm, and tapers towards the apex, where the apical cap at 6.80 mm is the thinnest region. Every segment thickens from ED to ES, and the thickening is graded along the long axis: the apical segments thicken by 52% to 64%, the mid-cavity segments by 51% to 64% and the basal segments by 37% to 61%. Over the normal group the mean thickness rises from 7.99 mm to 12.21 mm, a thickening of 52.9%.

Table 4.8. Regional AHA 17-segment wall thickness on the reconstructed geometry for the normal and HCM groups.

		Normal (NOR, $n = 20$)			**Hypertrophic (HCM, $n = 10$)**		
#	**Segment**	**ED (mm)**	**ES (mm)**	**Δ (%)**	**ED (mm)**	**ES (mm)**	**Δ (%)**
1	Basal anterior	8.72	12.25	+40.4	12.54	16.62	+32.6
2	Basal anteroseptal	8.81	12.10	+37.4	12.83	16.02	+24.9
3	Basal inferoseptal	8.68	11.96	+37.8	12.32	16.35	+32.8
4	Basal inferior	8.16	12.53	+53.6	10.96	16.37	+49.4
5	Basal inferolateral	7.94	12.81	+61.3	10.91	17.19	+57.5
6	Basal anterolateral	8.38	12.12	+44.6	11.83	17.37	+46.9
7	Mid anterior	7.94	12.10	+52.5	11.70	17.16	+46.6
8	Mid anteroseptal	8.55	12.98	+51.7	13.99	18.10	+29.4
9	Mid inferoseptal	8.57	12.98	+51.4	13.43	19.17	+42.8
10	Mid inferior	7.93	12.46	+57.1	11.89	18.32	+54.1
11	Mid inferolateral	7.96	13.02	+63.5	11.35	18.84	+66.0
12	Mid anterolateral	7.92	12.39	+56.4	11.13	17.73	+59.3
13	Apical anterior	7.28	11.75	+61.4	10.96	17.40	+58.7

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Table 4.8. AHA 17-segment thickness: NOR versus HCM *(continued).*

		Normal (NOR, $n = 20$)			Hypertrophic (HCM, $n = 10$)		
#	**Segment**	**ED (mm)**	**ES (mm)**	**Δ (%)**	**ED (mm)**	**ES (mm)**	**Δ (%)**
14	Apical septal	7.49	12.00	+60.2	11.33	17.59	+55.3
15	Apical inferior	7.28	11.72	+61.1	10.76	16.43	+52.6
16	Apical lateral	7.33	12.03	+64.0	10.61	17.28	+62.9
17	Apex	6.80	10.32	+51.8	9.67	14.71	+52.1
Overall		**7.99**	**12.21**	**+52.9**	**11.68**	**17.23**	**+47.5**

Note. Values are group means measured with the Laplace field. Δ is the relative thickening from ED to ES. These measurements summarise the proposed model output reconstructed from sparse SAX contours and are not an independent anatomical reference.

HCM thickness exceeds NOR thickness in every segment, with the largest differences in the mid-cavity anteroseptal and inferoseptal segments, where the HCM group reaches 13.99 and 13.43 mm compared with 8.55 and 8.57 mm in the normal group. Table 4.9 evaluates this separation at the patient level rather than treating the segments as independent observations. Each patient contributes one value per outcome: the mean and the largest of the ED segment means, and the relative ED-to-ES increase, so group differences in thickening are in percentage points. The confidence intervals come from resampling patients within each group [63], and Hedges' g [64] expresses the group difference relative to the patient-to-patient spread.

Table 4.9. Patient-level comparisons between the normal and HCM groups (mean ± SD). Differences are HCM minus NOR with 95% percentile-bootstrap CIs from 10 000 resamples [63]; positive Hedges' g [64] indicates a larger HCM value. Bold CIs exclude zero.

Measure	**NOR**	**HCM**	**Difference (95% CI)**	**Hedges' g**
Mean ED AHA-17 thickness (mm)	7.99 ± 0.88	11.66 ± 1.66	**+3.67 [2.54, 4.67]**	3.01
Maximum ED segment (mm)	9.07 ± 0.98	14.44 ± 1.94	**+5.37 [4.11, 6.56]**	3.82
Systolic thickening (percentage points)	53.2 ± 11.6	47.9 ± 11.5	−5.3 [−13.9, 3.0]	−0.45

Both ED confidence intervals exclude zero, and both effect sizes are large. The interval for relative systolic thickening includes zero, so this cohort shows no clear group difference in proportional thickening. None of the normal cases and three of the ten HCM cases have a maximum segment mean above the 15 mm clinical reference line [3].

Figure 4.5 shows the patient-level distributions of absolute ED thickness and relative systolic thickening.

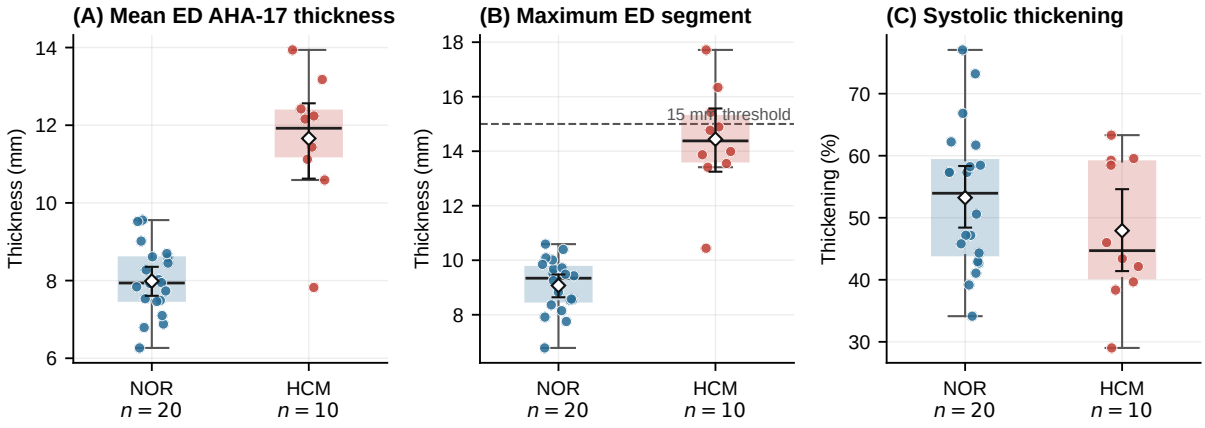

Figure 4.5. Patient-level AHA-17 wall-thickness outcomes for the normal and HCM groups. The dashed line marks the 15 mm clinical reference for maximal ED LV wall thickness; the plotted maximum segment mean is not diagnostically equivalent to it.

The two groups overlap more in relative thickening than in absolute thickness.

4.4. Discussion

The research questions of Section 1.3 are addressed through contour fidelity, agreement between methods, and internal consistency, not anatomical accuracy, because no independent 3D reference is available (Section 4.1).

RQ1 compares the learned reconstruction with RBF and SSM reconstructions of the same contours. Closure does not separate the methods: every surface is watertight after the shared repair stage, so closure is a property of the pipeline rather than of the learned prior. The relevant differences are contour fidelity and the geometry inferred between slices.

The proposed model follows its input rings to within about one in-plane pixel (Table 4.1). Expert contours contain annotation errors and are restricted to the pixel grid, so a surface that matches them to 0.13 mm, as the RBF fit does by construction, also reproduces those errors. A pixel-sized deviation may reflect smoothing rather than missed observations. The SSM fit has a similar mean distance but a wider spread between patients: its distance increases markedly on HCM cases, whereas the model distance changes little. An SSM can represent only shapes within the variation of its training population, and thick hypertrophic walls lie partly outside it. The proposed model, which is conditioned directly on the contours, appears less tied to an average shape.

The pairwise comparison shows where the three reconstructions disagree (Table 4.2). On the endocardium they differ by about one in-plane pixel, far less than the 10 mm slice spacing, and cavity DSC is high for every pair. Myocardial agreement is clearly lower, and myocardial volume varies by several tens of millilitres between methods. The contours fix the cavity outline

at each slice, but both surfaces must be inferred between slices, and small changes in the inferred epicardium alter myocardial volume and thickness much more than cavity overlap. Lower myocardial agreement therefore shows that the wall geometry depends on the reconstruction prior; it does not show that one method is anatomically better. Because wall thickness is measured on this geometry, the spread between methods is a lower bound on the uncertainty of every thickness value in this chapter. The proposed model lies closest to the RBF fit, but part of this agreement is expected, because the training targets were produced by a similar surface fit (Section 4.1.2).

Slice removal degrades the three methods by similar amounts (Table 4.3). The proposed model has the largest epicardial displacement with three slices but not the largest change in myocardial volume, so no method is more robust on every metric. These values describe stability relative to each method's own complete-stack surface, not anatomical error.

RQ2 asks whether the proposed phase-conditioned model can reconstruct paired surfaces at ED and ES while keeping a positive wall thickness. The architecture fixes the order of the two surfaces, because the epicardial field is defined as the endocardial field minus a strictly positive offset. On the output, the Laplace-derived thickness increases in all seventeen AHA segments between ED and ES (Table 4.7), which is the expected direction of systolic wall thickening. The evaluation did not measure the minimum pointwise separation between the extracted meshes, test for surface intersections after repair, or report the ED–ES change in cavity volume, and no fitting baseline is available at ES. Positive separation holds in the continuous fields by construction, but it has not been verified at every point of the extracted meshes, and accuracy at ES remains untested.

RQ3 asks how wall thickness depends on the reconstruction prior and on the measurement method, and whether regional and pathology-related differences are retained.

With the method fixed, the reconstruction prior mainly shifts the absolute scale (Table 4.5). Thickness on the proposed reconstruction and on the RBF fit is almost perfectly correlated across patients, yet values from the proposed model are consistently higher. The SSM fit changes both the scale and, to a lesser extent, the ranking, as expected from a stronger shape prior. A high correlation therefore does not imply agreement in millimetres; the bias reveals the offset.

With the geometry fixed, the measurement path matters (Table 4.6). Yezzi–Prince and the EDT boundary sum agree closely with the Laplace field, whereas the SDF cone-ray method is both biased and highly variable. A likely geometric explanation is that straight rays leave the curved wall too early or too late, whereas the Laplace and Yezzi–Prince paths follow the curvature of the wall and the EDT boundary sum does not depend on a single direction. The two effects are of similar size: the cone-ray bias of +3.31 mm is close to the 2.53 to 2.91 mm offset

between the proposed-model and SSM geometries with the non-ray methods. Thickness values are therefore comparable only when both the reconstruction and the measurement method are the same.

At the patient level, the HCM group is thicker at ED in both mean thickness and thickest segment, with confidence intervals that exclude zero and large effect sizes (Table 4.9). Relative systolic thickening does not separate the groups in this cohort. With ten HCM patients, these are descriptive differences within the proposed model output rather than evidence of diagnostic ability.

4.5. Limitations of the Evaluation

Without an independent 3D reference, contour fidelity and agreement between reconstructions measure consistency, not anatomical accuracy. The unobserved geometry between slices also affects wall thickness, so absolute values remain dependent on the reconstruction prior. Expert manual thickness measurements are unavailable, and neither clinical use nor replacement of established measurement methods has been validated.

The general evaluation uses held-out ACDC and M&Ms-2 patients, but the pathology analysis includes only 30 ACDC cases, 20 normal and 10 with HCM, limiting clinical generalisation. The seventeen segments from each patient are repeated measures, not independent samples. Relative thickening is sensitive to differences in wall sampling because ED thickness appears in the denominator. Larger cohorts and independent anatomical references are needed to confirm these findings.

Measurements depend on segmentation quality, AHA orientation, and inference-grid resolution, while acquisition differences across datasets may also affect the output. Only the heatmap in Figure 4.3 is scaled to the input segmentation; the tables report thickness measured directly on the meshes. The synthetic epicardium is constructed rather than measured and may still influence the output after clinical fine-tuning.

Slice-removal stability (Table 4.3) may reflect a prior that responds weakly to observations rather than better reconstruction. Removal levels also use different patient subsets, with the greatest removal limited to the longest stacks. Comparisons cover RBF and SSM fitting, not spline lofting, Poisson reconstruction, or other learned models. No ablations isolate synthetic pre-training, local conditioning, phase conditioning, or positive surface coupling, so their individual contributions and the need for the full architecture remain untested.

CHAPTER 5

Conclusions

This chapter summarises the work and answers the three research questions of Section 1.3 on the evidence of Chapter 4. It then states the contributions, limitations, and directions for future work.

5.1. Summary of the Work

Wall thickness is measured on the three-dimensional LV, but SAX cardiac MRI records separate slices with an in-plane pixel size below 2 mm and several millimetres between them [1, 46]. The wall between slices must therefore be reconstructed. A method was developed to recover both ventricular surfaces from these contours and measure wall thickness beyond the acquired slices.

The proposed 3D reconstruction model represents the LV surfaces as continuous signed-distance fields [17]. It was first pre-trained on synthetic shapes sampled from an SSM [12] to learn the general ventricular form, then fine-tuned on ED and ES surfaces fitted to ACDC [1] and M&Ms-2 [22] segmentations to capture differences between patients and cardiac phases.

The proposedmodel combines a global PointNet pathway [38], which summarises the overall shape, with a local feature grid [53], which retains nearby contour information. Cardiac phase is supplied as input so that one network reconstructs both ED and ES. The epicardial field is defined by subtracting a strictly positive learned offset from the predicted endocardial signed-distance field, preventing crossing in the continuous representation. The fields are then converted into meshes for measurement.

The reconstruction was compared with an RBF implicit fit [16] and an SSM fit to the same rings, examining contour fidelity, agreement between methods, and sensitivity to slice removal. Without an independent three-dimensional reference, these comparisons describe consistency rather than establish which surface is closest to the true anatomy.

Myocardial wall thickness was measured on each geometry with four local methods [42, 43, 44, 45]. Comparisons across geometries and within the same geometry separated the effects of reconstruction and measurement. Surface maps and AHA 17-segment summaries [11] showed regional thickness, changes from ED to ES, and differences between normal and HCM [3] patients.

5.2. Answers to the Research Questions

RQ1 compared the learned reconstruction with RBF and SSM fits to the same contours. The proposed model follows its input to within about one in-plane pixel (1.50 mm on the endocardium and 1.96 mm on the epicardium), lies closer to the RBF fit than to the SSM fit, and, unlike the SSM, keeps a similar contour distance on normal and HCM cases. The three methods agree on the cavity but not on the wall. The learned reconstruction performs comparably to conventional fitting and handles abnormal anatomy more evenly than the SSM, but is not more robust to slice removal, which degrades all three by similar amounts.

RQ2 asked whether the proposed phase-conditioned model can reconstruct paired endocardial and epicardial surfaces at ED and ES while maintaining a positive wall thickness. The answer is a qualified yes. One network reconstructs both phases, the epicardial field cannot cross the endocardial field by construction, and the reconstructed wall thickens in every AHA segment, from 9.21 mm at ED to 13.87 mm at ES. Pointwise separation on the extracted meshes and accuracy at ES were not verified.

RQ3 asked how strongly wall thickness depends on the reconstruction prior and the measurement method, and whether regional and pathology-related differences are retained. Both factors matter. The prior mainly shifts the absolute scale while largely preserving the ranking of patients, and a straight-ray measurement departs strongly from methods that follow the curved wall. Absolute thickness values are therefore comparable only within the same reconstruction and measurement pipeline. The reconstructed geometry does retain group differences: the HCM group is 3.67 mm thicker than the normal group in patient-level mean ED thickness, whereas relative systolic thickening does not separate the groups. With ten HCM patients, this is a descriptive result rather than a diagnostic one.

5.3. Contributions

This work makes three contributions. First, a shared data pipeline combines synthetic SSM meshes with clinical ED and ES segmentations from two public datasets. It stores contour rings, surface samples, signed-distance targets, and wall-offset targets in one format for both pre-training and fine-tuning. Second, the model processes endocardial and epicardial contours separately, combines global shape and local contour information, and uses a supervised positive offset to keep the epicardial field outside the endocardial field. Third, an evaluation approach applicable to other networks assesses contour fidelity, agreement between reconstructions, and sensitivity to slice removal. It also compares thickness methods on the same geometry, with AHA summaries and patient-group comparisons.

5.4. Limitations

The limitations in Section 4.5 bound these answers. Without an independent 3D reference, geometric results measure contour fidelity and agreement between methods, while thickness results measure reproducibility rather than accuracy. The pathology cohort is small, and no ablation experiment isolates the contribution of individual model components.

5.5. Future Work

These limitations set the order of the next steps. The first need is an anatomical reference. Higher-resolution volumetric acquisitions, or expert three-dimensional surface and wall-thickness annotations on a larger external cohort, would turn the agreement figures reported here into accuracy figures and would show which of the three priors lies closest to the true wall. The second is a wider and fully matched comparison, adding spline lofting [48], Poisson reconstruction [65] and learned alternatives [18, 20] under the same patients, physical spacing, repair procedure and metrics, with error maps at the apex, base and most curved parts of the wall. Controlled tests should then remove synthetic pre-training, local conditioning, phase conditioning and the positive-offset coupling one at a time to isolate their contributions. Pointwise separation and intersection checks on extracted meshes would address the remaining geometric checks in RQ2. Replacing the binary phase label with a continuous position in the cardiac cycle would allow coherence in time to be tested alongside continuity in space.

In sum, the proposed model turns separated SAX contours into closed, paired LV surfaces that honour the observations, thicken consistently from diastole to systole, and support local wall-thickness analysis. Disagreement between equally plausible reconstructions nevertheless shows how much of the wall remains inferred rather than observed, and how carefully its thickness must be interpreted.

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CHAPTER A

Supplementary Methodological Details

This appendix summarises the training data streams and partitions, RBF-derived clinical supervision meshes, training-loss terms, and model configuration described in Chapter 3.

A.1. Training Data Streams

Synthetic meshes establish a geometric shape prior during pre-training; clinical MRI data support fine-tuning on end-diastolic and end-systolic anatomy.

Table A.1. Training data streams and their roles.

Stream	Phase	Source	Training role
Synthetic SSM	ED	UK Digital Heart Project LV SSM	Pre-training to establish a geometric shape prior
Real MRI	ED	ACDC and M&Ms-2	Fine-tuning on patient-specific diastolic anatomy
Real MRI	ES	ACDC and M&Ms-2	Fine-tuning on systolic anatomy with phase conditioning

A.2. Dataset Composition and Partitions

Clinical samples are divided into training, validation, and test partitions, while accepted synthetic meshes are reserved for pre-training.

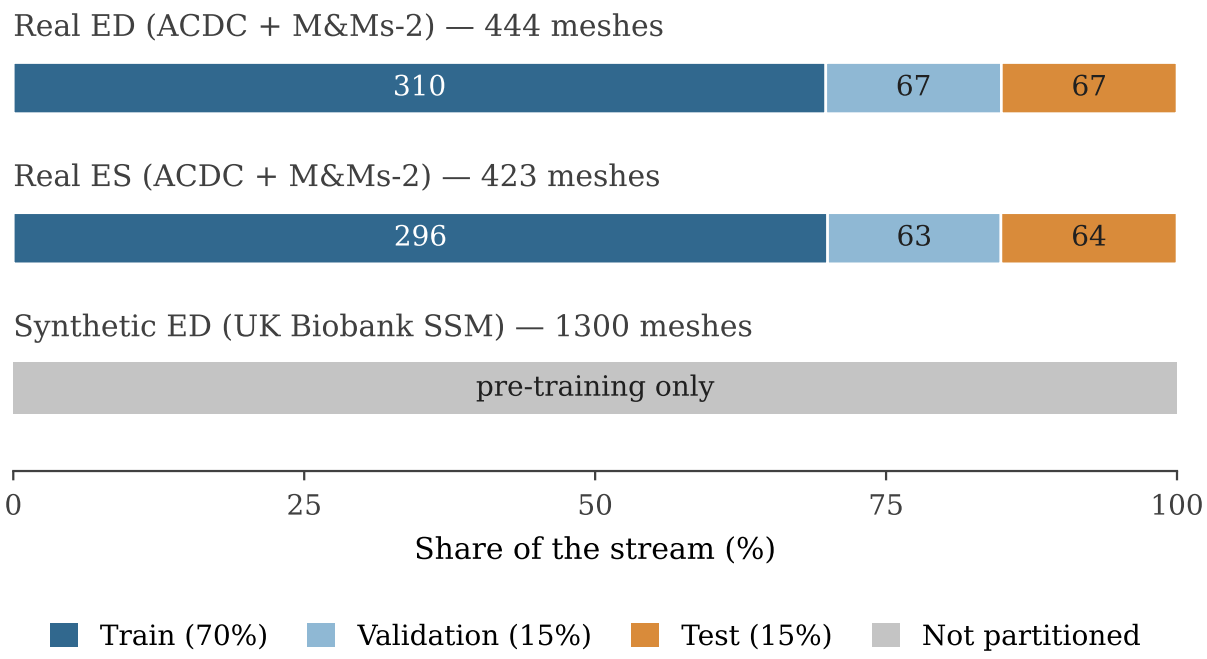

Figure A.1. Valid clinical samples divided into the 70 %/15 %/15 % partitions. The 1300 accepted synthetic meshes are reserved for pre-training.

A.3. Clinical Mesh Preparation

The clinical supervision meshes are constructed from SAX contours using RBF interpolation, surface extraction, and geometric validation.

Table A.2. Operations used to construct the RBF-derived clinical supervision meshes.

Operation	Purpose
RAS+ canonicalisation	Standardises patient-coordinate orientation while preserving physical spacing
Label harmonisation	Maps ACDC and M&Ms-2 labels to common endocardial and epicardial definitions
Largest-ring extraction and resampling	Retains 60 ordered points per boundary on each acquired SAX slice
In-plane normal estimation	Defines an outward direction at every contour point
Off-surface constraints at ± 2.5 mm	Establishes the sign and local scale of each implicit field
Thin-plate-spline RBF fit	Interpolates a continuous field with degree 1 and regularisation parameter 0.5
Evaluation on a 1.0 mm grid	Discretises the continuous field for surface extraction
Signed-distance regularisation ($\sigma = 1.2$ mm)	Reduces voxel-scale corrugation before surface extraction
Connected-region filtering and marching cubes	Removes invalid regions and extracts the zero level set
Topology validation and conditional correction	Preserves valid RBF surfaces and corrects only failed shells
Geometric plausibility checks	Rejects failed fits and invalid endocardial–epicardial pairs

A.4. Training-Loss Terms

The five loss terms combine surface and volumetric supervision with signed-distance regularisation and constraints on the decoder wall offset.

Table A.3. Purpose of the five training-loss terms.

Term	Purpose	Effect on reconstruction
$\mathcal{L}_{\text{surf}}$	Fits zero-level sets to surface samples	Places endocardial and epicardial surfaces on the target meshes
$\mathcal{L}_{\text{SDF}}$	Fits cached query-point distances	Provides dense volumetric supervision
$\mathcal{L}_{\text{eik}}$	Encourages $\|\nabla f\| = 1$	Regularises the decoder as a signed-distance field
$\mathcal{L}_{\text{off}}$	Repels off-surface points from zero	Reduces spurious zero crossings away from the myocardium
$\mathcal{L}_{\text{WT}}$	Fits the predicted wall offset to wall targets	Constrains separation of the predicted fields

A.5. Model Configuration

The configuration below specifies the contour conditioning, decoder architecture, output fields, and inference grid used by the model.

Table A.4. Main architecture settings of the model.

Component	Configuration
Input features	3D coordinates, tissue label, and cardiac phase
Global encoder type	Shared network with endocardial and epicardial max-pooling
Local conditioning	Coarse 3D contour feature volume with trilinear interpolation
Latent dimension	256
Local volume size	16^3 with 32 channels
Positional encoding	Fourier features with $L = 3$ frequency bands
Decoder	8-layer network, width 512, skip connection at layer 4
Activation	Softplus (smooth)
Local injections	Local feature paths into the decoder trunk and offset head
Output heads	Endocardial signed distance and positive decoder wall-offset field
Offset mapping	Sigmoid component with $\tau_{\text{min}} = 0.05$ and $\delta_{\text{cap}} = 0.45$, plus learned soft-hinge headroom
Inference grid	96^3 query grid for marching cubes

A.6. Training Configuration

The settings below specify the optimisation, schedule, batching, and hardware used to train the model in both stages.

Table A.5. Training configuration used for the model.

Setting	Value
Training strategy	SSM pre-training, then fine-tuning on real ED/ES
Training mode	Combined ED and ES with phase conditioning
Augmentation	Contour-input perturbations for synthetic ED and real ED/ES; targets unchanged
Optimiser	AdamW
Learning rate	2×10^{-5}
Scheduler	Cosine annealing with warm restarts ($T_0 = 100$ epochs)
Weight decay	5×10^{-4}
Training duration	Approximately 200 pre-training epochs, then 776 fine-tuning epochs (best checkpoint: 695)
Batch size	8 per GPU $\times$ 2 GPUs, accumulated for 2 steps (effective batch size 32)
Gradient clipping	1.0
Precision	Automatic mixed precision
Hardware	2 GPUs in parallel
Inference grid	96^3